

The Cost of Bank Regulatory Capital

Matthew C. Plosser

Federal Reserve Bank of New York, USA

João A. C. Santos

Federal Reserve Bank of New York, USA and Nova School of Business and Economics, Portugal

Basel I introduced capital requirements for undrawn commitments, but only for revolvers with an original maturity greater than one year. We use this regulatory discontinuity to estimate the impact of capital regulation on the cost and composition of credit. Following Basel I, short-term commitment fees declined relative to long-term commitments and issuance of short-term facilities increased. Our results highlight the sensitivity of credit provision to capital regulation, particularly for banks with less capital. We are able to infer that low-capital banks are willing to forego twice as much income from fees to reduce required regulatory capital by a dollar. (*JEL* G21, G28)

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Modigliani and Miller (1958) demonstrated that in the world of Arrow-Debreu — where markets are complete, information is symmetric and other frictions are not present — a firm's value is independent of its capital structure. Of course, the assumptions of this setting are often violated. For example, the deductibility of interest expenses from income taxes makes debt financing relatively attractive to firms. On the other hand, the costs of financial distress make equity financing appealing. Firms trade-off these and other frictions when optimizing their capital structure.

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In addition to these considerations, banks also consider the presence of a government safety net and the unique role of deposits as both a source of financing and a service, which further tilts their capital structure toward debt financing. These forces explain why banks operate with higher leverage ratios than do nonfinancials (Pennacchi and Santos 2021) and are central to understanding bankers' claims that capital regulation is costly, prompting them to charge higher prices for their services. Empirically, it has been difficult to evaluate the claim that regulatory capital is costly. Estimating the cost of bank funding, particularly capital, presents several challenges: changes in capital regulations are rare, they are often endogenous to evolving risks, and they typically affect all banks, thus precluding a well-identified counterfactual.

We overcome these challenges by capitalizing on a unique opportunity provided by the transition to the Basel I Accord. Prior to that Accord, undrawn commitments (formal standby facilities and credit lines) were not subject to capital requirements in the United States. Basel I set specific capital requirements for these commitments, but only for those with a maturity exceeding one year. If a commitment was undrawn and had an original maturity less than 365 days, it did not contribute to risk-weighted assets and as a result did not necessitate that banks set aside capital. This regulatory change introduced a discontinuity that had important effects on the marketplace for credit lines — prior to the change there was scant evidence of 364-day facilities in the marketplace, but as illustrated in Figure 1 these instruments became prevalent following Basel I and their share of credit facility originations grew. The Basel II Accord sought to reduce the maturity discontinuity by extending capital standards to short-term undrawn commitments; however, Basel II did not completely eliminate the capital advantages of 364-day facilities and, unlike Basel I, Basel II was not uniformly applied across banks and was never fully adopted by the United States.

Our approach exploits the regulatory discontinuity to estimate the profits banks are willing to sacrifice so that they can avoid holding regulatory capital. We start by outlining a framework of the bank's pricing decision that implies a relation between the relative pricing of commitments and the level of capital held by the bank. The framework motivates our difference-in-differences empirical strategy in which we compare banks' pricing of undrawn commitments with maturities below one year and above one year around the implementation of Basel I. Our results imply that banks are willing to charge lower fees for commitments that bypass capital requirements.

Building on this insight, we investigate whether low-capital banks are willing to forego more profits than their peers with larger capital buffers. We perform a set of robustness and placebo tests to ensure our findings are in fact driven by the Basel I Accord. In addition, we investigate whether there was an increase in the relative quantity of short-term credit lines in the years immediately after Basel I. Lastly, we study banks' relative pricing of short-term commitments following the publishing of Basel II to investigate whether the

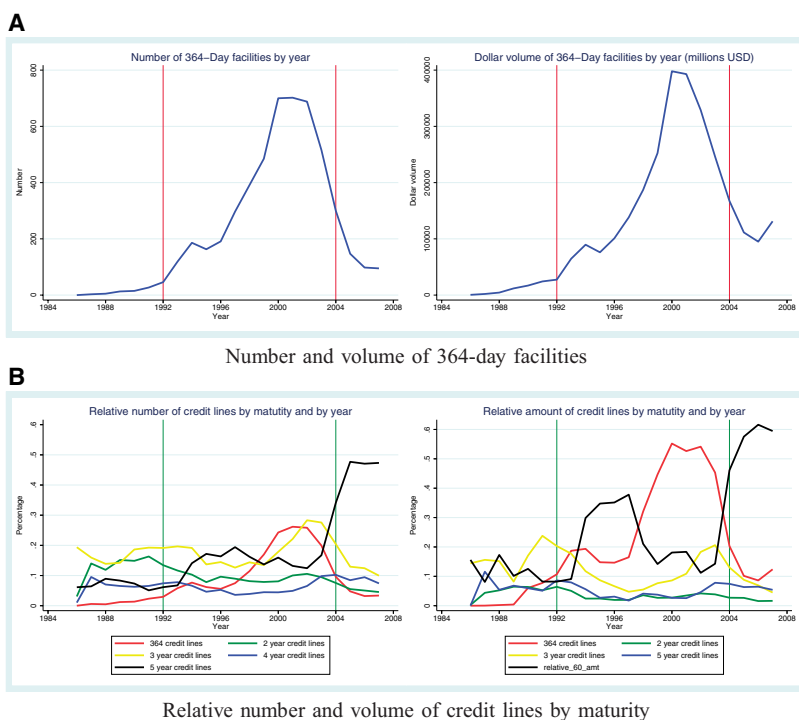


Figure 1
Credit line origination activity by year

This figure depicts the time series of the number (left) and dollar volume (right) of 364-day facility originations. Panel B plots the time series of the relative number and dollar volume of originations for 364-day facilities, and 2-, 3-, 4- and 5-year credit lines as a proportion of total credit facilities originated in a given year.

new regulatory regime attenuated the Basel I results. Throughout our analysis we include controls for loan-, borrower-, and bank-specific factors, as well as market conditions known to affect commitment pricing.

The pricing structure of a commitment includes an undrawn fee and an all-in-drawn spread. The undrawn fee is comprised of a commitment fee and an annual fee that the borrower must pay for funds committed, but not drawn.¹ The undrawn fee compensates the bank for the liquidity risk it incurs by guaranteeing the firm access to discretionary funding over the life of the credit line. In contrast, the all-in-drawn spread, which is defined over Libor and is equal to the annual cost to a borrower for drawn funds, compensates the bank for the credit risk it incurs when the borrower draws down on its credit line.

¹ DealScan uses the wording all-in-undrawn spread when referring to the price firms pay on undrawn commitments, but in reality that price is not a spread because the fees are not markups over market interest rates.

The Basel Accord's special treatment of short-term commitments only applies to the undrawn portion of the facility. Once the borrower draws down its commitment, the utilized amount receives a capital treatment that is independent from its maturity. Given this, we expect the impact of the Basel I Accord to be concentrated in undrawn fees, and we focus our investigation accordingly. We nonetheless complement the study of undrawn fees with an exploration of all-in-drawn spreads.

We find that commitments with maturities up to one year, including 364-day facilities, become relatively less expensive following the passage of Basel I. Consistent with our priors, we find that the undrawn fees on short-term commitments decline relative to those of commitments with maturities longer than one year. This finding is robust to different control specifications, to variation in the event window, and to alternative samples of commitments. We also obtain this result in several additional robustness tests.

The relative decline in the undrawn fees of short-term commitments appears to be driven by the change in the capital regulation. The decline is greater for low-capital banks, which have stronger incentives to loosen their capital constraints. In a placebo test, we estimate the impact of a pricing discontinuity at a 2-year maturity, rather than at 1 year, and find no relative change in undrawn fees at the time of Basel I; this is consistent with Basel I shifting prices rather than any concurrent change in the pricing of maturity. We also conduct a placebo test that compares the relative pricing of term loans of varying maturity around the Basel I Accord and do not find an effect, indicating that the change in undrawn fees is related to the specific regulations applied to commitments. Further, our evidence on the relative decline in the undrawn fees of short-term commitments is robust to various fixed effects, including bank-year, bank and borrower, and even bank-borrower. Our investigation of commitment pricing around Basel II suggests that the issuance of short-term commitments declines, but the pattern in undrawn fees only partially reverses, consistent with the fact a smaller set of short-term commitments qualified for capital relief.

Finally, we do not find evidence of a similar relative decline in the all-in-drawn spreads of short-term credit lines. This is expected: all-in-drawn spreads summarize the cost of a drawn commitments, which are treated the same under Basel I regardless of maturity.

Based on our findings, we estimate that the average bank is willing to pay almost \$0.02 annually to reduce regulatory capital by one dollar, while the average low-capital bank is willing to pay twice as much. This suggests that the cost of regulatory capital is lower for many banks than they have indicated. However, these estimates may be a lower bound and banks may be willing to pay more. The inferred cost of regulatory capital might be low because banks have market power over their borrowers that limits pass-through of the regulatory changes to prices. Also, firms may be extremely sensitive to changes in commitment pricing thereby exhausting the relevant market opportunity for

undrawn credit lines despite banks' willingness to lower fees even further. As we document, the small decline in undrawn fees induces a large shift in the relative amount of short-term commitments, highlighting the sensitivity of credit to the regulatory capital regimes.

The shift toward short-term credit lines made it easier for banks to monitor borrowers and to manage liquidity risk, but it exposed borrowers to additional refinancing risk and additional repricing risk. It also effectively made liquidity insurance cheaper, which not only could increase investment but also could raise risk-taking. The results highlight that capital standards affect not only the quantity of lending, but also the loan terms, leading to an increase in refinancing risk for borrowers.

Our paper is related to the literature that attempts to infer the importance of bank capital by studying banks' responses to changes in capital requirements. A segment of this literature, including [Bernanke and Lown \(1991\)](#), [Hancock and Wilcox \(1993, 1998\)](#), [Haubrich and Wachtel \(1993\)](#), [Berger and Udell \(1994\)](#), [Peek and Rosengren \(1995\)](#) and [Beatty and Gron \(2001\)](#), attempts to ascertain whether Basel I affected the volume of bank lending. More recently, [Gropp et al. \(2019\)](#) and [Jiménez et al. \(2017\)](#) also study bank responses to regulatory capital changes but they build on the 2011 European Banking Authority capital exercise and the Spanish provisioning requirements, respectively. In contrast to this body of research, we focus on the loan-level response, conditional on the bank and, in some specifications, the bank-borrower pair. This helps mitigate bank and borrower factors that might otherwise explain our findings. [Benetton et al. \(2021\)](#) and [Behn, Haselmann, and Wachtel \(2016\)](#) consider the implementation of internal ratings-based capital requirements under Basel II, showing they affect loan prices and volumes. Our setting and analysis are unique in that we consider the capital held for undrawn commitments and detect price effects in addition to a large change in the maturity structure of credit lines.

In addition, we infer the cost of regulatory capital by exploiting a loophole in the corresponding regulation. [Kisin and Manela \(2016\)](#) use the cost of holding assets in an asset-backed commercial paper (ABCP) conduit to estimate the marginal cost for which banks are indifferent to creating a zero-capital requirement investment. The authors assume that banks can move their assets into an ABCP conduit, so that if they are indifferent, the marginal cost of adding to the ABCP conduit must be equivalent to the benefit of not holding capital against the investment. Their identification strategy is sensitive to two assumptions: (a) banks can move any asset into an ABCP conduit; (b) these contributions can be financed at low CP rates. Deviations in either assumption can significantly change the inferred cost of capital to banks.

We rely on a different setting and alternative assumptions, both of which result in a novel estimate. Our approach uses a difference-in-differences empirical strategy that compares the relative cost of credit lines over time. Therefore, it is critical to control for changes in the composition of borrowers

and financial conditions in order to ensure that our estimates are well-identified. We do so using a linear model, loan-, borrower-, and bank-specific controls as well as various fixed effects. In addition, we assume perfect competition and that there is no constraint on the demand for credit lines. Implications for violating these assumptions are discussed in Section 6.

Our motivating framework implies that low-capital banks, with smaller buffers, should be more willing to forgo earnings in exchange for capital. We document that low-capital banks, those in the bottom quartile of the equity to assets ratio distribution, indeed are willing to sacrifice more earnings in exchange for capital relief. This is consistent with [Benetton et al. \(2021\)](#), who document that U.K. capital-constrained banks displayed a higher willingness to pay for capital during the Great Financial Crisis.

Two other related papers are [Kashyap, Stein, and Hanson \(2010\)](#) and [Van den Heuvel \(2008\)](#). The former paper estimates the impact of higher capital requirements on loan rates based on the substitution away from tax deductible debt. The latter paper estimates the cost of bank capital requirements for consumer welfare using a general equilibrium model in which capital requirements reduce liquidity services.

1. Background on Basel Accords

Prior to the 1980s, banks in the United States were not subject to specific capital standards.² The federal banking agencies — the Federal Reserve, the Office of the Comptroller of the Currency (OCC), and the Federal Deposit Insurance Corporation (FDIC) — first introduced numerical regulatory capital requirements in 1981. The three agencies required banks to meet a minimum leverage ratio of capital to total assets, despite differences across agencies both in what constituted capital and the minimum requirement.³ However, these differences persisted only until 1985, when the agencies agreed to set the minimum primary capital ratio at 5.5% of total assets. This followed the passage of the International Lending and Supervision Act of 1983, which directed the federal banking agencies to issue regulations addressing capital adequacy.

Soon after the 1985 harmonization, concerns began to emerge that the existing capital ratio did not differentiate bank assets by riskiness and did not factor in the risk exposures associated with innovative and growing banking activities, most notably off-balance-sheet activities at larger institutions. Additionally, there were growing concerns that differences in capital standards across countries were affecting banks' ability to compete internationally. These

² See [Burhouse et al. \(2003\)](#) for a detailed review of the history of bank capital regulation in the United States.

³ The Fed and the OCC announced a minimum primary capital adequacy ratio of 6% for community banks and 5% for larger regional institutions. The FDIC established a threshold capital-to-assets ratio of 6% and a minimum ratio of 5%.

concerns led to negotiations among the central bank governors of the Group of Ten (G-10) countries that culminated in the 1988 approval of the Basel I Accord.

The Basel I Accord assigned a risk weight for each on-balance-sheet exposure and specified the minimum capital banks must hold against their risk-weighted assets. Risk weights ranged from 0 to 100%, depending on the creditworthiness of the counterparty and the nature of the risk.⁴ For example, on-balance-sheet exposures to corporate borrowers received a 100% weight.

The Accord also specified a credit conversion factor for off-balance-sheet exposures (e.g., credit commitments), which defined the amount of capital the bank had to set-aside depending on the maturity of the commitment and the underlying risk weight. Commitments to corporations with an original maturity in excess of one year were treated as off-balance-sheet exposures and the undrawn portion of the commitment was assigned a 50% conversion factor. In contrast, commitments with an original maturity of up to one year, or the ability to be unconditionally canceled at any time, received a 0% conversion factor.⁵ The conversion factor difference required banks to set aside capital when they extended commitments with a maturity greater than one year, but not when they extended shorter-maturity commitments. As a consequence, banks began to issue so-called “364-day” facilities.

To the extent that equity capital is costly for banks, the difference in conversion factors made short-term credit lines (those with maturities less than one year at origination) relatively less expensive following the introduction of Basel I. This gives rise to the primary hypothesis we consider in this paper:

Hypothesis: The relative cost of short-term to long-term credit lines declined after the introduction of Basel I when compared to the period prior to the Basel Accord.

The Basel II Accord, which was finalized in June of 2004, sought to lessen the preferential treatment of facilities with an original maturity of up to one year. Basel II introduced two alternative approaches for banks to determine the amount of required capital for credit exposures, the standardized approach and the internal ratings based approach. Under the standardized approach, the required capital set aside for a credit exposure is based on external credit agency ratings. In contrast, under the internal approach, banks use their internal rating systems to ascertain the credit risk, and by extension the capital requirement, of their exposures.

Both approaches changed the treatment of short-term facilities. Under the standardized approach, these facilities only benefit from a 0% credit conversion

⁴ See Santos (2001) for a detailed description of the Basel I Accord.

⁵ The 0% risk conversion applied only to the portion of the commitment that was undrawn. Once drawn, that portion would be treated as an on-balance-sheet exposure.

factor if the bank has the discretion to unconditionally cancel the facility at any time without prior notice, or if the facility contains a covenant triggering automatic cancellation in case of a deterioration in the borrower's financial condition. Any short-term facility that does not meet these revised criteria is subject to a 20% credit conversion factor.

For banks using the internal ratings approach, two variations are possible. Under the "foundation" internal ratings approach, short-term facilities are subject to a conversion factor of up to 75%, unless the facility is unconditionally cancellable without prior notice, in which case it will qualify for a 0% conversion factor. Banks that adopt the advanced internal ratings approach have the discretion to estimate the potential exposure at default and set the credit conversion factor for each facility.

Whichever approach banks use, Basel II increased the likelihood that banks would need to hold capital against short-term facilities. Note, however, that Basel II did not fully reverse the advantages of these facilities. Not only are some short-term commitments still able to receive a 0% conversion factor but other short-term loans receive an advantaged conversion factor of 20%. Hence, the cost reversal may not be of the same magnitude as the decrease predicted in the first hypothesis, although the quantity of these lines should decrease as the qualifying terms are less attractive to borrowers. Nevertheless, we evaluate the introduction of Basel II to confirm the importance of capital regulation on the issuance of short-term lines and their relative cost.

However, the Basel II exercise bears an important caveat. Unlike Basel I, where the timing of adoption and the relative capital benefits were clear, Basel II advanced with much less clarity. The United States was one of the key proponents of Basel II, and federal bank regulators put out an advance notice of proposed rulemaking (ANPR) in August 2003 (following the third Basel II consultative paper finalized in June 2003), signalling their intent to adopt Basel II. However, the United States did not ultimately adopt Basel II until December 2007, and by then it only implemented the advanced internal ratings-based approach for the large banks, retaining Basel I for the remaining banks. While the degree of delay was not anticipated, the timing of these events makes it less clear when U.S. banks chose to incorporate the Basel II framework into their capital management and loan-pricing decisions.

2. Methodology

We choose an empirical methodology designed to estimate the impact of capital regulations on the market for credit lines and to recover the economic value of capital to banks. To motivate our methodology, we first outline a simple representation of the bank's decision, which suggests a difference-in-differences empirical approach.

2.1 The bank's problem

Assume the bank maximizes a profit function, which depends on a revenue function, R , a cost function, C , and a preference for a capital buffer, U . Revenue is a function of loans, L (which include loans and other risky on-balance-sheet investments), undrawn credit facilities, F , and borrowed funds, B (including deposits). Undrawn facilities can be either short-term, F^S , or long-term, F^L . The cost function depends on loans, L , credit facilities, F , equity, E , and borrowing, B .

The utility function for the capital buffer, U , is such that the excess capital is valued in units comparable to net revenues (i.e., dollars), and is concave ($U' > 0, U'' < 0$) in the capital buffer, which is defined as the distance between the bank's actual equity capital, E , and the minimum regulatory capital, $\bar{E}$. The required capital of the bank is a function of its lending and long-term undrawn facilities; hence, the capital buffer can be written: $\Omega = E - \bar{E}(L, F^L)$. The required capital of the bank does not depend on short-term undrawn facilities after the Basel I Accord (more on this below).

We opted for relying on this capital buffer approach, rather than imposing an inequality constraint on capital, to capture the empirical observation that banks manage to a capital buffer (Berger et al. 2008) and to ensure regulatory capital always has some marginal value, albeit one that is strictly decreasing with the buffer size. Also, it provides for a more plausible and nuanced approach to capital management than a linear constraint that results in a bang-bang solution. Nevertheless, when we discuss our estimates on the cost of regulatory capital we consider the implications of a linear regulatory constraint as well as a world where the bank faces multiple capital-related incentives.

We simplify the analysis by abstracting from the mix of risky assets and the optimal size of the bank because neither of these directly involve the choice between originating short- or long-term credit facilities. As a result, banks choose their mix of credit facilities and funding quantities to maximize profits, Π , and their utility from the capital buffer, U , relative to total assets and subject to a balance sheet constraint:⁶

$$\max_{L, F^S, F^L, E, B} \Pi = R(L, F^S, F^L, B) - C(L, F^S, F^L, E, B) + U(\Omega(E, L, F^L))$$

$$\text{s.t. } L = B + E.$$

Substituting and solving for the first order condition with respect to lending yields:

$$\frac{\partial \Pi}{\partial L} = (R_L - C_L) - (R_B - C_B) = U' \bar{E}_L. \quad (1)$$

From Equation 1, we can see that the marginal profits from lending net of marginal financing costs are equal to the marginal utility from the change

⁶ In other words, the bank is solving for lending and funding quantities relative to total assets. Introducing alternative classes of on-balance-sheet assets does not affect our findings.

in capital buffer. This implies that growing lending, all else equal, increases the bank's required regulatory capital thereby decreasing the available buffer ($\bar{E}_L > 0$). In this instance, U' reflects the cost of a change in required regulatory capital to the bank with respect to what it is willing to forego in terms of earnings.

In the context of our natural experiment, the capital buffer is not a function of credit line maturity in the pre-Basel I period. As we discussed in the previous section, in the pre-Basel I era capital regulation did not factor in to the issuance of undrawn commitments. Therefore, when we difference the first-order conditions on credit lines we find:

$$(R_{FL} - R_{FS}) - (C_{FL} - C_{FS}) = 0. \quad (2)$$

The difference-in-marginal revenue from undrawn lines, the undrawn fees, is equal to the difference in the costs and has no implications for capital. After Basel I the risk-weighted capital requirement is sensitive to long-term credit lines ($\bar{E}_{FL} > 0$), but not short-term lines ($\bar{E}_{FS} = 0$), hence differencing the two respective FOCs yields

$$(R_{FL} - R_{FS}) - (C_{FL} - C_{FS}) = U' \bar{E}_{FL}. \quad (3)$$

Empirically, we observe undrawn fees which correspond to the marginal revenue from issuing an undrawn line; however, we do not observe the associated costs. Consequently, we cannot recover the marginal utility of the capital buffer. To estimate the left-hand side, we proxy for the costs in the post-Basel period using the relation implied in Equation (2) which equates the difference in costs to the difference in prices (undrawn fees). This suggests a difference-in-differences empirical specification which recovers the incremental pricing charged on long-term lines and implies the value of the capital buffer to the bank:

$$(R_{FL} - R_{FS})_{Post} - (R_{FL} - R_{FS})_{Pre} = U' \bar{E}_{FL}. \quad (4)$$

The final formulation contains a sufficient number of empirically observable terms, including the differential pricing of long- and short-term lines over time and the marginal impact of a long-term line on the required regulatory capital ($\bar{E}_{FL}$). This formulation allows us to recover the marginal utility of capital to the bank. Of course, the pricing of credit lines depends on other factors not captured here that can vary over time, across banks, and across borrowers. For this reason, and as we explain in the next section, we carefully consider a broad set of empirical controls that can influence credit-line pricing. The estimates we recover can then be used in Section 6 to estimate the value of capital to banks. At that time, we will revisit the underlying assumptions of the model and their implications for interpreting the results.

2.2 Empirical methods

Equation (4) motivates our use of a difference-in-differences (DiD) empirical approach to estimate the change in pricing related to capital regulation.

Towards that end, we begin by estimating the following model on credit lines originated around Basel I:

$$\begin{aligned} \text{UNDRAWN}_{f,l,b,t} &= c + \alpha ST_{f,l,b,t} + \beta \text{BASEL1}_t + \gamma \text{BASEL1}_t \times ST_{f,l,b,t} \\ &+ \sum_{i=1}^I \psi_i X_{i,l,t} + \sum_{j=1}^J \nu_j Y_{j,f,t-1} + \sum_{k=1}^K \eta_k Z_{k,b,t-1} + \rho M_t + \epsilon_{f,t}. \end{aligned} \quad (5)$$

$\text{UNDRAWN}_{f,l,b,t}$ is the undrawn per annum fee on credit line l of firm f from bank b at issue date t . According to DealScan, the undrawn fee includes both the commitment fee and the annual fee that the borrower must pay its bank for undrawn funds committed under the credit line. This contrasts with the all-in-drawn spread, which is expressed as a spread over the benchmark London interbank offering rate (Libor), and is defined as the annual amount the borrower pays on drawn commitments. The all-in-drawn spread takes into account both the annual fee associated with the loan and the credit spread. Because the differential treatment granted by the Basel I Accord to commitments with maturities up to one year applies only to funds committed, but not yet drawn, we focus our investigation on undrawn fees.

ST is a dummy variable equal to one for credit lines with a maturity at origination up to (and including) one year. In some specifications we narrow this definition to include only facilities with maturities up to 11 months and in other specifications we consider only 364-day facilities.⁷ BASEL1 is a dummy variable equal to one for credit lines originated after the Basel I Accord. U.S. banks were required to apply Basel I on a transitional basis starting in 1991, but the Accord became fully phased in starting in 1993. We begin our investigation on a sample containing credit lines originated between 1987 and 2003, with the post Basel I period defined by the years 1993–2003. We start in 1987 and end in 2003 because the Basel II Accord was finalized in 2004. We also consider a balanced, 6-year window (1990–1995, maintaining 1993 as the first year after Basel I) to reduce concerns that we are comparing disparate time periods.

The key variable in our pricing model is the interaction between ST and BASEL1 . The coefficient for this variable, γ , estimates the change in the relative price of commitments with maturities up to one year versus longer-term commitments from the period prior to the Basel Accord I compared to the post period. If capital is indeed costly to banks, then this coefficient should be negative. Ideally, one would like to compare the pricing of short-term commitments to commitments with maturities only slightly above one year. However, the number of observations to carry out this exercise is not

⁷ Each of these variants assures us that the target commitments are below the one-year cutoff specified in the Basel I Accord, but they pose some challenges. For example, very few facilities are classified as 364-day prior to Basel I, because they did not have any special regulatory status at that time.

sufficient, because maturities at origination are typically issued at discrete annual horizons. Instead, we consider alternative samples that rely on a tighter control group comprised of commitments with maturities between 1 and 3 years.

We choose control variables that are likely to influence banks credit lines' pricing policies over time.⁸ To that end, we include loan-, borrower-, and bank-specific controls as well as market conditions at the time of origination. These controls accommodate changes in loan terms, firm characteristics, and bank characteristics that might otherwise confound identification. In robustness tests, we go one step further and allow the control coefficients to vary pre- and post-Basel I to allow for changes in pricing determinants over time.

Our loan-specific controls ($X_{i,l,t}$) include loan amount, the number of lenders in the syndicate, along with indicators to control for whether the loan is senior, secured, the presence of dividend restrictions, the presence of a guarantor, and the loan purpose. Our firm-specific controls ($Y_{j,f,t-1}$) are only available for a subset of borrowers, therefore, we consider specifications both with and without these variables. Our controls include standard variables, such as firm size (proxied by sales), leverage, profitability, asset tangibility, and market-to-book ratio, along with cash flow variables (net working capital and the log of the interest coverage truncated at zero). We complement these variables with two equity controls, the stock return (in excess of the market return) and volatility of the firm's stock return. We also include indicator variables for each credit rating and single-digit SIC code category. In robustness tests, we consider specifications with borrower fixed effects to account for otherwise unobserved changes in the composition of borrowers. Shifts in borrower demand that happen to coincide with Basel I are difficult to control for, but given we observe a *decline* in pricing and an *increase* in usage, the results are consistent with an increase in the supply of short-term facilities by banks rather than an unrelated boom in borrower demand.

Our bank-specific controls ($Z_{k,b,t-1}$) focus on the characteristics of the lead arranger bank. Our reasoning is that the lead bank negotiates not only the initial loan terms but also is the institution charged with enforcing the terms, so its characteristics will most affect fee pricing behavior. Our bank-specific controls include bank size, profitability, risk, liquid asset holdings and subordinated debt (both scaled by assets), and credit rating, along with the capital-to-assets ratio. To further reduce concerns as to whether our results are bank driven, we also consider specifications with bank fixed effects.

Our overall market controls (M_t) include the spread between BBB- and AAA-rated bond index yields at the time of the loan origination. When we investigate all-in-drawn spreads, we include the term premium as measured by the difference between yields on 10-year and 3-month treasuries at the time of

⁸ See Bord and Santos (2014) and Santos and Viswanathan (2020) for empirical studies of commitments' undrawn fees.

the credit line origination. Although syndicated loans are typically a floating rate, their credit spreads could possibly vary with the term premium. Each control is defined in the [Internet Appendix](#).

We estimate our model using a pooled regression, first excluding firm controls to broaden the sample. Next, we add firm controls. Finally, and while still using firm controls, we include bank fixed effects. Throughout, our errors are clustered two ways, by borrower and time (year-quarter) to account for persistent errors within a borrower and cross-sectional correlations in prices (e.g., [Fama 1998](#)).

2.2.1 Addressing confounding explanations Although the focus of our analysis is on undrawn fees, a potential concern with our base specification is that the estimated impact could be driven by general changes in the pricing of loan maturity rather than by Basel I. We address this concern by conducting two distinct placebo tests. The first test compares the undrawn fees on credit lines with maturities between 1 and 2 years to the undrawn fees on credit lines with maturities above 2 years around the implementation of Basel I. The second test builds on all-in-drawn spreads. As we noted in Section 2, we would expect Basel I to affect only undrawn fees and not credit spreads on drawn credit lines. This is because once drawn, credit lines receive the same treatment under Basel I regardless of their maturity. Also, we would not expect Basel I to affect the relative credit spreads of term loans because Basel I treatment of term loans did not vary with their maturity. We investigate these two assertions using a model similar to Equation (5) but with the all-in-drawn spread on the left-hand side.

In addition to the placebo tests, we carry out a test based on the Basel II Accord. The Basel II Accord sought to reverse, although only partially, the special treatment that Basel I afforded short-term facilities. However, as we discussed in Section 2, even though the United States put out an advance notice of proposed rulemaking in August 2003, clearly signalling their intent at the time to adopt Basel II, the United States only approved final rules for implementing Basel II in November 2007. Therefore, it is possible that U.S. banks did not adjust their loan policies in the wake of the international approval of Basel II in June of 2004. Notwithstanding this uncertainty, we investigate undrawn fees on credit lines around Basel II. We restrict our analysis to the period 2000-2007 and specify the years between 2005 and 2007 as the post Basel II period. Similar to our analysis of Basel I, we consider a tighter sample period (2002-2007), and restrict the control group to credit lines with maturities up to 3 years. In addition, we explore a variety of fixed effects specifications.

3. Data

The data for this project come from several sources. We use the Loan Pricing Corporation's (LPC) DealScan database of business loans to identify credit lines originated by banks. The DealScan database contains information on

credit line terms, including the undrawn fee, the all-in-drawn spread over Libor, the maturity, the seniority status, and the stated purpose; the borrower, including its sector of activity, and its legal status (private or public firm); and finally, the lending syndicate, including the identity and role of banks in the loan syndicate at origination.

While DealScan begins in the 1980s, it is not comprehensive in the early part of the decade. Thus, we begin our sample in 1987. We end our sample in 2007, before the Great Financial Crisis, because of the ample evidence that it significantly affected bank lending and could obscure any impact of capital regulation.⁹

We use the Center for Research on Securities Prices's (CRSP) stock prices database to link companies and subsidiaries that are part of the same firm, and to link companies over time that went through mergers, acquisitions or name changes. We then use these links to merge the LPC and Compustat databases in order to obtain the financial condition of firms at the time they borrow. We also use CRSP to determine each publicly listed borrower's excess stock return, and stock return volatility. Financial information is typically only available for publicly listed firms, but we will consider alternative empirical specifications that exclude these controls to confirm our results hold for both private and public firms.

We rely on the Salomon Brothers/Citigroup yield indexes on new long-term industrial bonds to control for changes in the market's credit risk premium. We calculate the yield difference between the indexes of triple-B and triple-A rated bonds as these indexes go back to December of 1988. We complement these indices with Moodys' corporate seasoned bond yields in order to get information on the triple-B spread as far back as January of 1987.

Finally, we use the Reports of Condition and Income compiled by the FDIC, the Comptroller of the Currency, and the Federal Reserve System to obtain bank data for the lead bank(s) in each loan syndicate. Whenever possible we focus on the consolidated bank holding company financials from Y9C Reports. If these reports are not available, then we rely on Call Reports, which contain bank-level data.

3.1 Sample characteristics

Table 1 presents the characteristics of the samples that we use to investigate Basel I (left) and Basel II (right), respectively. The left columns compare credit lines issued before Basel I (1987-1992) with those issued afterward (1993-2003). The right panel, in turn, compares credit lines issued before Basel II (2000-2004) with those issued afterward (2005-2007).

⁹ Examples of the impact of bank-specific conditions on corporate lending during the crisis include Santos (2011), who focuses on the impact of banks' financial condition, and Ivashina and Scharfstein (2010) and Cornett et al. (2011), both of whom focus on the impact of banks' exposure to unused credit lines.

Table 1
Sample characteristics

Variables	Basel I sample			Basel II sample		
	Bef	Aft	Diff	Bef	Aft	Diff
<i>A. Undrawn fees</i>						
ST	38.54	16.94	-21.61***	16.336	11.747	-4.590***
LT	37.91	31.93	-5.98***	32.934	22.325	-10.609***
ALL	37.99	27.90	-10.09***	27.184	21.542	-5.642***
<i>B. Loan controls</i>						
LAMOUNT	3.821	4.821	1.000***	5.241	5.763	0.521***
LMATURITY	1.123	0.914	-0.210***	0.809	1.403	0.594***
SECURED	0.430	0.434	0.003	0.388	0.413	0.025**
DIVIDEND	0.003	0.506	0.503***	0.534	0.617	0.082***
GUARANTOR	0.000	0.044	0.044***	0.107	0.128	0.021***
SPONSOR	0.027	0.040	0.013***	0.030	0.052	0.023***
CORPUSES	0.334	0.234	-0.100***	0.271	0.462	0.191***
DEBTREPAY	0.194	0.294	0.100***	0.126	0.021	-0.105***
WORKCAPITAL	0.241	0.148	-0.093***	0.252	0.325	0.073***
CPBKUP	0.018	0.134	0.116***	0.226	0.040	-0.186***
M&A	0.132	0.114	-0.018**	0.059	0.087	0.028***
LENDERS	6.314	9.080	2.766***	10.460	10.994	0.535***
<i>C. Firm controls</i>						
LSALES	5.695	6.710	1.015***	7.380	7.742	0.362***
LEVERAGE	0.330	0.302	-0.029***	0.306	0.270	-0.036***
MKTBOOK	1.452	1.772	0.320***	1.713	1.825	0.112***
PROFMARGIN	0.015	0.011	-0.004	0.012	0.057	0.045***
NWC	2.289	3.903	1.614***	3.858	5.446	1.588***
LINTCOV	1.720	2.090	0.370***	2.110	2.432	0.322***
TANGIBLES	0.795	0.743	-0.052***	0.737	0.710	-0.027***
R&D	0.015	0.018	0.003**	0.019	0.017	-0.002*
ADVERTISING	0.014	0.010	-0.004***	0.010	0.011	0.001
STOCKRET	0.001	0.00	-0.00***	0.001	0.000	-0.000***
STOCKVOL	0.033	0.033	-.000	0.032	0.019	-0.013***
AAA	0.001	0.007	0.006***	0.010	0.013	0.003
AA	0.009	0.034	0.025***	0.038	0.020	-0.018***
A	0.073	0.149	0.076***	0.199	0.151	-0.048***
BBB	0.111	0.165	0.054***	0.235	0.258	0.023**
BB	0.099	0.104	0.004	0.109	0.172	0.063***
B	0.091	0.057	-0.034***	0.055	0.071	0.016***
CCC	0.006	0.001	-0.005***	0.001	0.003	0.002*
CC	0.004	0.002	-0.002**	0.001	0.000	-0.000
<i>D. Bank controls</i>						
LASSETS	3.468	5.215	1.747***	5.951	6.649	0.699***
SUBDEBT	0.010	0.023	0.013***	0.024	0.021	-0.003***
ROA	0.001	0.003	0.002***	0.003	0.003	0.000***
CHARGEOFFS	0.002	0.001	-0.001***	0.001	0.001	-0.000***
LIQUIDITY	0.230	0.198	-0.032***	0.191	0.161	-0.031***
CAPITAL	0.058	0.073	0.015***	0.074	0.082	0.008***
ROAVOL	0.003	0.001	-0.002***	0.001	0.001	-0.001***
<i>E. Firm controls</i>						
BBBSPREAD	1.044	0.994	-0.050***	1.369	.752	-.617***
TREASURYSLOPE	1.802	1.571	-0.231***	2.126	0.441	-1.685***
Observations	2,133	10,452		5,577	3,015	

This table characterizes the samples we use in our investigation of credit lines' undrawn fees around Basel I (left panel) and Basel II (right panel), respectively. See [Internet Appendix](#) for the definitions of all the variables reported in the table. The sample period used in the left panel is 1987-2004, with 1993 being the first year after Basel I. The sample period used in the right panel is 2000-2007, with 2005 being the first year after Basel II. For each variable, we report the average before and after the regulatory change as well as the difference. * $p < .1$; ** $p < .05$; *** $p < .01$.

We compare credit lines using a wide set of variables. Panel A compares undrawn fees for credit lines based on maturity type. Panels B, C, and D compare sets of loan-, borrower- and bank-specific controls that we use in our investigation of pricing, respectively. Finally, panel E compares our control for market conditions, the triple-B bond spread and the treasury term premium at the time of the credit line origination.

Looking at panel A, we see that undrawn fees decline after Basel I and Basel II. However, undrawn fees of short-term commitments, regardless of how we identify them, decline by more than for long-term commitments in the post-Basel I period, consistent with Hypothesis 1. In contrast, we see that undrawn fees of short-term commitments decline by less than those of long-term commitments during the post-Basel II period, which is consistent with our expectation that the Basel II Accord removed some of the favorable treatment that Basel I had given to short-term commitments.

Turning our attention to the remaining panels, we see that many of the controls exhibit statistically significant differences over time. In the interest of space, we do not provide a detailed analysis of these differences. However, they suggest the importance of investigating the robustness of our findings to a specification that allows control variables to have different coefficients before and after each Accord. Further, one control variable, maturity, is worth examining because it provides insight into our hypothesized effect of the Basel Accords. The average maturity declines significantly after the introduction of Basel I (from 4 years to 3 years), and reverses after Basel II (from 3 years to 4 years). These changes are consistent with our expectation that Basel I incentivized the origination of commitments with maturities below one year, while Basel II, at least in part, reduced that incentive.

To gain a deeper understanding of the evolution in credit line maturity, in Table 2 we report transition matrices for credit line maturities around Basel I (top panel) and around Basel II (bottom panel). The table reports the distribution of commitment maturity *after* the Accord to the most recent commitment maturity by the same borrower *prior* to the Accord. Both of these transition matrices report information only for borrowers that take out commitments before *and* after the Accord.

The top panel depicts two results that support our assertion that Basel I made commitments with maturities up to one year relatively more attractive. First, looking at the diagonal of the matrix, which focuses on borrowers that retained the maturity of their commitments before and after Basel I, we see that borrowers who took out 1-year commitments before Basel I are the most likely to take out 1-year commitments afterward. Second, looking at the first column, we see that there was a high incidence of borrowers that switch to one-year commitments after Basel I. For example, among borrowers that originally took 2-year commitments, we see that nearly as many of them switch to 1-year commitments (17.4%) when compared to those that continue to take out 2-year commitments after Basel I (17.6%). As further evidence of the increase

Table 2
Transition matrices of loan maturities

A. Transition matrix around Basel I

Maturity before	Maturity after Basel I									
	1	2	3	4	5	6	7	8	9	10
1	25.7	16.3	21.6	8.9	13.9	4.6	1.5	0.4	0.2	6.8
2	17.4	17.6	25.7	8.9	13.4	5.7	2.1	0.9	0.0	8.5
3	19.1	11.4	24.0	10.2	19.6	5.2	3.1	0.4	0.0	7.0
4	15.7	10.3	19.0	13.0	24.1	5.7	3.3	0.5	0.3	8.1
5	16.0	10.0	17.3	7.9	24.6	8.9	5.2	1.4	0.4	8.5
6	15.4	7.0	18.7	12.1	22.4	8.2	5.7	2.4	0.0	8.2
7	17.0	7.7	16.2	8.8	21.9	8.0	9.8	2.3	0.5	7.7
8	11.2	5.9	14.4	6.4	22.5	15.5	4.8	5.9	3.7	9.6
9	17.7	3.2	11.3	9.7	21.0	6.5	6.5	6.5	11.3	6.5
10	21.6	4.2	13.5	8.4	17.7	6.7	9.2	4.2	6.7	7.6
All	18.0	11.2	20.3	9.6	19.7	6.9	4.3	1.5	0.7	7.8

B. Transition matrix around Basel II

Maturity before	Maturity after Basel II									
	1	2	3	4	5	6	7	8	9	10
1	25.8	7.8	11.6	4.5	45.6	3.6	0.7	0.0	0.0	0.5
2	9.8	12.1	20.9	12.1	38.1	4.2	0.5	0.5	0.0	1.9
3	6.3	6.3	19.1	10.1	50.8	5.7	0.6	0.0	0.0	1.2
4	3.7	6.2	11.6	15.1	51.7	8.7	0.7	0.0	0.0	2.2
5	4.2	1.9	5.9	9.6	60.4	14.2	1.8	0.1	0.0	1.8
6	2.4	0.6	4.7	10.1	39.1	33.7	5.9	1.2	0.0	2.4
7	2.1	2.1	6.4	2.1	27.7	27.7	27.7	2.1	0.0	2.1
8	0.0	0.0	4.2	8.3	37.5	20.8	20.8	4.2	0.0	4.2
9	0.0	0.0	0.0	0.0	0.0	50.0	50.0	0.0	0.0	0.0
10	1.0	3.1	8.3	13.4	46.4	23.7	2.1	0.0	0.0	2.1
All	7.3	4.7	11.4	9.9	51.9	11.1	1.9	0.2	0.0	1.6

This table reports the transition matrices for borrowers that took out credit lines before and after Basel I (top panel), and borrowers that took out credit lines before and after Basel II (bottom panel). The sample period used in the top panel is 1987-2004, with 1993 being the first year after Basel I. The sample period used in the bottom panel is 2000-2007, with 2005 being the first year after Basel II. In each panel we keep all of the credit lines taken out after the Basel Accord and compare their maturities with the borrower's last credit line before the Accord. Rows should add to 100, except for rounding errors.

in the attractiveness of 1-year commitments after Basel I, the first column in the top panel is always larger than the first column in the bottom panel. In other words, for each maturity the percentage of borrowers that switched to 1-year commitments after Basel I is always higher than the percentage of borrowers that does a similar switch after Basel II.

One final insight from the bottom panel of Table 2 is that there was a large percentage of borrowers that switched from short-term commitments prior to Basel II to 5-year commitments (the most popular long-term maturity) afterward. For example, 45.6% of the borrowers that took out a 1-year commitment prior to Basel II choose a 5-year commitment afterward. For borrowers that took out 2- and 3-year commitments the corresponding figures are 38.1% and 50.8%. For reference, these percentages around Basel I were 13.9%, 13.4%, and 19.6%, respectively. This suggests that a large set of borrowers that use to rely on a funding strategy based on the rollover of short-term commitments choose to switch to long-term funding around the time of

Basel II. We will come back to this issue when we investigate banks' credit line pricing around Basel II.

4. Basel I Results

We begin by examining the time series of the undrawn fees on credit lines to verify that the difference-in-differences methodology is appropriate. To interpret the result of the DiD estimation as related to the differential treatment by capital regulation, we must assume that in the absence of Basel I the difference in fees between the short- and long-term credit lines is not explained by other trends over time. Figure 2 plots undrawn fees for both sets of credit lines indexed to 100 in 1992. Prior to Basel I, there is a stable relation between the two groups and no clear trend, but significant gaps emerge post Basel I that persist through the early 2000's. This supports the assertion that regulatory capital is costly, as the pricing of the short-term lines precipitously declines in tandem with Basel I, while long-term fees remain relatively elevated.

Of course, this insight is based on univariate comparisons and does not control for any of the factors known to help explain undrawn fees. We go one step further in our investigation of pre-trends and condition on control variables in order to better account for other factors that may affect fees over time. To do so, we estimate Equation 5, but rather than interact the short-term dummy with a Basel I indicator we interact it with a range of annual fixed effects.

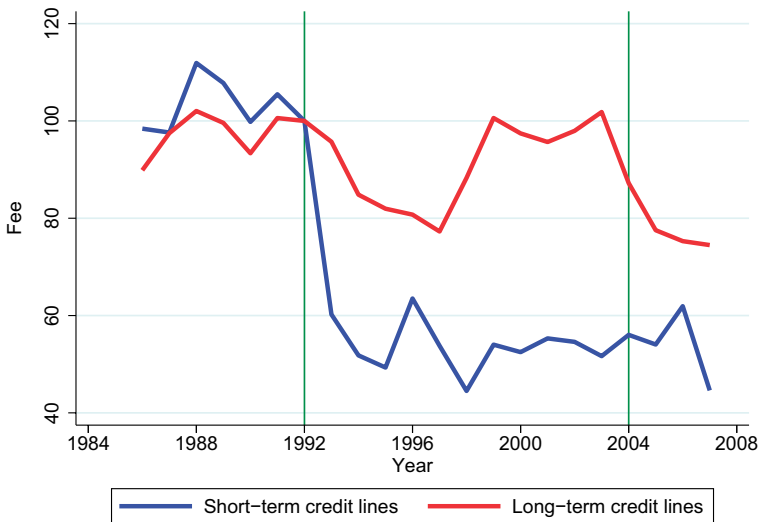


Figure 2
Undrawn fees around Basel I
This figure plots the average annual undrawn fees on short- versus long-term credit lines around Basel I. Fees are scaled to 100 in 1992.

We include the full set of controls (as in Table 3) and rely on the tighter control group of credit lines with maturity less than 4 years (e.g., Table 4, panel A). Figure 3 plots the coefficients for models with and without bank fixed effects. In both cases, the differences between short- and long-term credit lines are statistically indistinguishable from zero in the pre-Basel I period with no clear trend. In the post-Basel I years, the difference is negative, significantly different from zero and again lacks a discernible trend. Together, Figures 2 and 3 rule out differential trends in commitment fees and suggest a significant change in undrawn revolver pricing around Basel I.

4.1 Base specification

We proceed with our investigation by estimating our pricing model, Equation 5, on the sample of credit lines taken out between 1987 and 2003. The results of this exercise are reported in Table 3. Model 1 reports the pooled regression estimates excluding firm-specific controls, which allows us to consider credit lines of both private and publicly listed borrowers. Model 2 adds our set of firm-specific controls that restricts our sample to credit lines of publicly listed borrowers. Finally, Model 3 reports the results estimated with bank fixed effects and firm-specific controls again using the sample of publicly listed borrowers.

A careful inspection of the three variables in Table 3 that are critical to our analysis, *ST*, *BASEL1*, and the interaction between these variables, reveals some important insights. First, the results do not vary substantially across the models. While there are some differences in statistical significance, the variables that retain their significance also retain their signs across the three models. Second, prior to Basel I, short-term commitments had lower undrawn fees than long-term commitments, but the difference was generally not statistically significant. Third, after Basel I, long-term commitments observed a significant decline in undrawn fees of 2.4–3.8 basis points (bps). Last, and most importantly for our purposes, undrawn fees on commitments up to one year declined by 2.4–4.1 bps relative to those of longer term commitments following the passage of Basel I. This evidence supports, from a statistical point of view, Hypothesis 1 that the favorable treatment of short-term commitments lowered their undrawn cost relative to longer-term facilities.

Looking at our loan-, borrower-, and bank-specific controls as well as our market controls, we see that statistically significant coefficients are generally consistent with expectations. Loan terms and firm characteristics associated with risk tend to increase commitment fee pricing, while loan size decreases pricing. The BBB spread is positively associated with commitment fees indicating a role for market risk in the pricing of even undrawn lines. In the interest of space, we do not provide a more detailed discussion of these controls here.

Table 3
Undrawn fees on credit lines around Basel I

Variables	(1)	(2)	(3)
BASEL1	-2.42*** (-3.09)	-3.75*** (-3.87)	-3.20*** (-2.89)
ST	-1.67 (-1.35)	-2.85*** (-2.74)	-1.98* (-1.93)
BASEL1 × ST	-4.10*** (-3.12)	-2.41** (-2.12)	-3.18*** (-2.81)
Loan controls:			
LAMOUNT	-1.91*** (-11.54)	-1.15*** (-4.53)	-1.25*** (-4.82)
RENEWAL	1.68 (0.99)	3.24* (1.76)	2.61 (1.60)
SECURED	11.73*** (29.35)	8.83*** (19.84)	8.80*** (18.15)
SECUREDMIS	4.51*** (12.83)	2.55*** (7.55)	2.27*** (7.01)
DIVIDEND	1.10*** (2.64)	1.44*** (3.60)	1.56*** (4.36)
GUARANTOR	1.07 (1.12)	0.73 (0.79)	0.68 (0.75)
SPONSOR	7.61*** (13.99)	5.85*** (6.14)	5.45*** (5.69)
CORPURPOSES	-5.18*** (-10.49)	-3.02*** (-4.85)	-2.94*** (-4.72)
DEBTREPAY	-4.47*** (-8.78)	-3.52*** (-6.04)	-3.38*** (-5.86)
WORKCAPITAL	-4.84*** (-8.20)	-2.30*** (-3.12)	-2.12*** (-3.00)
CPBACKUP	-9.60*** (-13.51)	-6.02*** (-8.28)	-5.94*** (-8.10)
M&A	-0.72 (-1.21)	1.87*** (2.73)	1.77** (2.56)
LENDERS	0.03** (2.13)	0.04** (2.24)	0.04** (2.19)
Firm controls:			
LSALES		-0.08 (-0.43)	-0.11 (-0.60)
LEVERAGE		4.23*** (3.44)	4.12*** (3.36)
MKTBOOK		-0.66*** (-2.67)	-0.71*** (-2.88)
PROFMARGIN		-5.45*** (-3.49)	-5.76*** (-3.74)
NWC		0.02** (2.04)	0.02* (1.89)
LINTCOV		-1.45*** (-6.27)	-1.35*** (-6.03)
TANGIBLES		-0.57 (-0.90)	-0.52 (-0.82)
R&D		-15.55*** (-3.50)	-13.35*** (-3.01)
ADVERTISING		4.15 (0.53)	1.61 (0.21)
STOCKRET		-277.45*** (-3.18)	-304.81*** (-3.50)
STOCKVOL		168.12*** (13.32)	169.63*** (12.59)
AAA	-14.74*** (-10.98)	-7.53*** (-5.95)	-7.93*** (-6.19)
AA	-13.51*** (-17.40)	-10.53*** (-10.46)	-10.46*** (-10.26)

(Continued)

Table 3
(Continued)

Variables	(1)	(2)	(3)
A	-10.88*** (-17.36)	-9.28*** (-12.24)	-9.33*** (-12.19)
BBB	-4.83*** (-7.68)	-4.52*** (-6.66)	-4.66*** (-7.00)
BB	4.12*** (6.59)	3.74*** (6.24)	3.49*** (5.80)
B	7.62*** (9.67)	5.74*** (5.98)	5.53*** (5.70)
CCC	13.79*** (4.62)	4.21 (1.44)	4.31 (1.52)
CC	12.82*** (3.54)	5.61 (1.37)	4.95 (1.19)
Bank controls:			
LASSETSbk	0.43*** (2.80)	0.32** (2.03)	0.42 (0.60)
SUBDEBTbk	-62.95*** (-2.88)	-53.04** (-2.43)	-63.85** (-2.45)
ROAbk	-87.76 (-0.85)	-148.20 (-1.40)	-130.85 (-1.22)
CHARGEOFFSbk	497.67*** (3.05)	383.01* (1.95)	47.08 (0.22)
LIQUIDITYbk	-5.99* (-2.34)	-3.66 (-1.40)	10.91*** (2.68)
CAPITALbk	-43.18*** (-3.99)	-34.42** (-2.56)	-5.19 (-0.32)
ROAVOLbk	399.76*** (3.27)	241.09* (1.68)	106.22 (0.70)
Market controls:			
BBBSPREAD	4.29*** (7.99)	2.75*** (5.73)	3.36*** (4.62)
Bank fixed effects	No	No	Yes
Observations	22,048	12,585	12,536
R-squared	.401	.491	.512

Models estimated on a sample of credit line originations taken out between 1987 and 2003. The dependent variable in each model is the annual undrawn fee on the credit line in basis points. All models include a set of indicators reflecting the borrower's single-digit SIC code. Model 1 excludes firm-level controls, Model 2 includes firm-level controls, and Model 3 includes firm-level controls and bank fixed effects. See the [Internet Appendix](#) for the definition of all the variables. Standard errors are clustered in two dimensions: borrower and time (year-quarter). * $p < .1$; ** $p < .05$; *** $p < .01$.

4.2 Tightening the Basel I tests

Our base models cast a broad net and compares credit lines with maturities at origination up to one year with all remaining credit lines. One concern with these models is that it includes a set of credit lines with a wide range of maturities that may not be comparable.¹⁰ To address this concern, we repeat our analysis using a restricted sample of credit lines with maturities less than or equal to 3 years.

The results of this test are reported in panel A of Table 4. In all three models, the coefficient for the interaction term, $BASEL1 \times ST$, is negative and

¹⁰ While the transition matrices in Table 2 suggest a high degree of substitution by borrowers between 1- and 5-year lines, pricing policies could vary more across these maturities.

Table 4
Basel I: Alternative samples and specifications for undrawn fees

<i>A. Tighter sample (maturity ≤ 3 years)</i>			
Variables	(1)	(2)	(3)
BASEL1	-2.43** (-2.48)	-3.16** (-2.57)	-1.74 (-1.28)
ST	-1.10 (-0.90)	-2.05* (-1.86)	-1.23 (-1.06)
BASEL1 $\times$ ST	-4.86*** (-3.63)	-4.39*** (-3.44)	-5.16*** (-3.84)
Bank fixed effects	No	No	Yes
Observations	12,052	7,250	7,202
R-squared	.382	.470	.497
<i>B. Tighter sample & time-varying coefficients</i>			
Variables	(1)	(2)	(3)
ST	-0.97 (-0.92)	-1.82* (-1.78)	-1.06 (-0.81)
BASEL1 $\times$ ST	-5.07*** (-4.07)	-4.60*** (-3.41)	-5.32*** (-3.40)
Bank fixed effects	No	No	Yes
Observations	12,052	7,250	7,202
R-squared	.387	.477	.503

The dependent variable in each model is the annual undrawn fee on the credit line in basis points. Models include the same controls as the corresponding models in Table 3 including industry indicator variables. Panel A reports the results when we restrict the sample to credit lines with maturities up to 3 years. Panel B reports results using the tighter control sample (maturity ≤ 3 years) and allowing for coefficients for controls to vary between the pre- and post-Basel I periods as in Table 4, panel B. Because of lack of interpretability we suppress the Basel I indicator coefficient in this panel. See Internet Appendix for the definition of all the variables. Standard errors are clustered in two dimensions: borrower and time (year-quarter). * $p < .1$; ** $p < .05$; *** $p < .01$.

statistically significant at the 1% level. In fact, restricting the control group to a more homogeneous set of credit lines increases the impact of Basel I on credit lines' pricing; short-term lines' undrawn fees now decline 4.4 - 5.2 bps more than longer term lines' fees following Basel I.

Our tests thus far account for a large set of loan-, borrower- and bank-specific controls as well as the market conditions at the time of issuance of commitments. Notwithstanding that, our specifications are not flexible enough to account for time variation in banks' loan pricing policies; in other words, coefficients for controls are assumed to be stable over time. To address this concern, we reestimate our models including interactions of each of our controls with the dummy variable identifying the Basel I period, *BASEL1*. The results of this exercise are reported in panel B of Table 4 and are cumulative with the sample change in the earlier panel. Comparing the results reported in Table 4 with the previous results we obtained without interacting all of our controls with *BASEL1*, we see that adding the new controls does not meaningfully affect our findings. Our key variable of interest, *BASEL1* $\times$ *ST*, continues to be negative and of similar magnitude in each of our models. Further, the interaction coefficient retains similar levels of statistical significance.

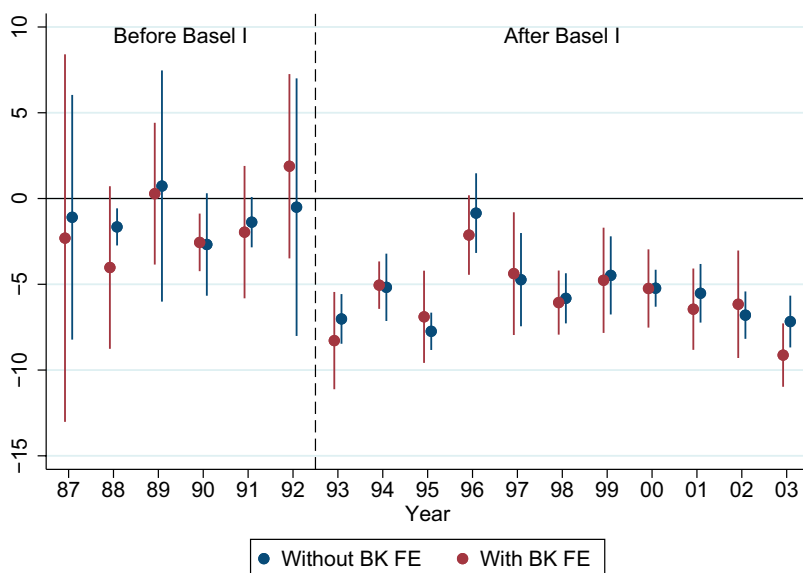


Figure 3

Conditional difference between short- and long-term undrawn fees

This figure plots the average annual difference between undrawn fees on short- versus long-term credit lines around Basel I estimated on the sample of credit lines with maturities up to 4 years conditional on the full set of controls in Table 3.

4.3 Bank heterogeneity

The results we have reported thus far estimate the average change in pricing across our sample of banks. However, we would expect the effects of the Basel I Accord to be larger for banks that have smaller capital buffers and are closer to their regulatory minima. In the context of the model in Section 2.1, banks obtain a decreasing marginal return from incremental capital buffer ($U'' < 0$) such that a bank with a large buffer values the incremental capital less than a bank with a small buffer, *ceteris paribus*. If this is indeed the case, Equation 4 implies that the differential in pricing will be greater for banks with lower capital buffers or, equivalently, lower capital ratios. A counter hypothesis would be that banks with less capital are less sensitive to buffers due to unobserved factors, such as less risk-averse management. In such cases, we would not find a relation between bank capital and the differential pricing of short term lines post-Basel I.

To investigate the importance of bank capital on its relative value, we classify banks as low capital if their equity-to-assets ratios are in the first quartile of the sample ratio (6.1%).¹¹ Next, we expand our empirical model to distinguish how

¹¹ We use the equity-to-assets ratio because we cannot observe their tier 1 risk-based ratio over the entire sample period.

low-capital banks adapt their pricing of short-term credit lines (relative to the remaining banks) to Basel I by interacting the capital constraint dummy with the DiD interaction terms. If low-capital banks value marginal improvements in capital their capital buffer more, then we would expect the coefficient for the low-capital indicator interacted with $BASEL1 \times ST$ to be negative, indicating that they reduce the relative price of short-term lines more than their better capitalized peers.

The results of this investigation are reported in Table 5. Looking at panel A of Table 5 we see that prior to Basel I low capital bank fees are not statistically different than other banks (see $LOWCAP \times ST$); however, our coefficient of interest, $BASEL1 \times LOWCAP \times ST$, is always negative and statistically significant. This is true even in Model 3 that includes bank fixed effects, suggesting that within a bank lower capital generates a greater pricing differential.

We can calculate the magnitude of the differential change in pricing. Well-capitalized banks experience a decline of 3.6–4.9 bps ($BASEL1 \times ST$); however, the low capitalization banks roughly double this as they are 5–4.8 bps lower. In total, low-capital banks are willing to price these lines roughly 8.4–8.6 bps lower than longer maturity lines versus low-capital banks in the pre-period ($BASEL1 \times LOWCAP + BASEL1 \times LOWCAP \times ST$).

Following our earlier exercises to tighten the Basel I tests, in panel B we use the tighter sample configuration, which restricts to credit lines with maturities up to 3 years and includes time-varying coefficients. These modifications to the sample and model increase the size and statistical significance of the differential pricing for low-capital banks. In this instance the total differential for low-capital banks pricing ranges from 8.8 to 9.2 bps.

A caveat to this analysis is that bank capital is endogenous. Low-capital banks may differ in other ways relative to high capital banks, including their risk attitude. For instance, low-capital banks may simply be more comfortable with their capital positions. In such a scenario, low-capital banks should not be more sensitive to the change in regulatory regime; therefore, this particular source of endogeneity biases coefficients toward zero. Also, we consider specifications with bank fixed effects, so if there are persistent differences in bank capital management, then the fixed effects should account for them. In sum, low-capital banks reduced undrawn fees by more than the remaining banks consistent with the idea that low-capital banks with smaller buffers place a higher value on capital relief.

5. Robustness Tests

In this section we report the results from a series of robustness tests to our key finding that the relative cost of short-term credit lines declined following the implementation of the Basel I Accord. First, we restrict the sample period to an event window of 6 years around the implementation of Basel I. Second, we

Table 5
Bank heterogeneity: Low-capital banks and undrawn fees around Basel I

Variables	A. Tighter sample (maturity ≤ 3 years)		
	(1)	(2)	(3)
BASEL1	-3.33*** (-2.83)	-3.70*** (-2.92)	-1.76 (-1.30)
ST	-1.82 (-0.92)	-2.11 (-1.23)	-0.81 (-0.59)
LOWCAP	-1.39 (-1.00)	-0.44 (-0.34)	0.82 (0.74)
BASEL1 $\times$ ST	-3.62* (-1.72)	-3.61** (-1.96)	-4.89*** (-3.15)
BASEL1 $\times$ LOWCAP	3.30** (2.32)	3.09** (2.15)	2.15** (2.09)
LOWCAP $\times$ ST	1.22 (0.51)	0.13 (0.06)	-0.69 (-0.38)
BASEL1 $\times$ LOWCAP $\times$ ST	-4.98** (-2.04)	-4.79* (-1.88)	-3.66* (-1.67)
Bank fixed effects	No	No	Yes
Observations	12,052	7,250	7,202
R-squared	.383	.471	.498

Variables	B. Tighter sample & time-varying coefficients		
	(1)	(2)	(3)
ST	-1.96 (-1.02)	-2.70* (-1.67)	-1.65 (-1.04)
LOWCAP	2.77* (1.65)	4.53*** (4.10)	4.62*** (2.94)
BASEL1 $\times$ ST	-3.59* (-1.77)	-3.04* (-1.74)	-4.06** (-2.35)
BASEL1 $\times$ LOWCAP	-1.36 (-0.73)	-2.50** (-2.14)	-2.07 (-1.40)
LOWCAP $\times$ ST	1.71 (0.75)	1.47 (0.75)	0.88 (0.45)
BASEL1 $\times$ LOWCAP $\times$ ST	-5.21** (-2.24)	-5.90*** (-2.77)	-5.13** (-2.42)
Bank fixed effects	No	No	Yes
Observations	12,052	7,250	7,202
R-squared	.388	.479	.505

The dependent variable in each model is the annual undrawn fee on the credit line in basis points. Models include the same controls as the corresponding models in Table 3 including industry indicator variables. *LOWCAP* is an indicator for loans originated by banks with an equity to assets ratio below the first quartile of the sample ratio (6.1%). Panel A reports results using the tighter control sample (maturity ≤ 3 years) as in Table 4, panel A. Panel B reports results using the tighter control sample (maturity ≤ 3 years) and allowing for coefficients for controls to vary between the pre- and post-Basel I periods as in Table 4, panel B. Because of lack of interpretability we suppress the Basel I indicator coefficient in this panel. See the [Internet Appendix](#) for the definition of all the variables. Standard errors are clustered in two dimensions: borrower and time (year-quarter). * $p < .1$; ** $p < .05$; *** $p < .01$.

use alternative ways to identify the short-term loans that benefit from Basel I. Third, we consider two placebo tests. Fourth, we consider a set of tests to account for unobservables. Finally, we examine the relative pricing of credit lines around the adoption of Basel II. We finish with a brief discussion of some additional robustness tests reserved for the [Internet Appendix](#).

5.1 Alternative sample window

While Figure 2 and time-varying control specifications suggest that the changes in pricing are persistent and relatively stable over the sample period (1987–2003), unobserved factors that vary over time could bias our results. At the expense of sample size, we explore a tighter event window, 1990–1995, to further establish the robustness of our key finding. The smaller, balanced sample period encompasses three years before (1990–1992) and after (1993–1995) the passage of Basel I. For this exercise we continue to limit the sample to credit lines with maturities less than or equal to 3 years.

The new estimates are reported in panel A of Table 6, which has a structure similar to prior tables. Narrowing the window around Basel I does not affect our key findings. The interaction term remains negative and highly statistically significant in each model. Further, narrowing the window of our test increases the magnitude of the $BASEL1 \times ST$ coefficient. Undrawn fees on short-term credit lines declined 4.7–6.2 bps relative to facilities with longer maturities.

5.2 Cost of 364-day facilities

All of the results reported thus far focus on commitments with maturities at origination up to and including one year. DealScan reports maturity in months, so we have categorized one-year facilities (e.g., 12 months) as short term because notwithstanding the reported months many of these facilities are in fact classified as 364-day facilities. These so-called “364-day” facilities appear to have been developed to benefit from the favorable treatment offered by Basel I.¹² However, as we noted above the Basel I discontinuity occurs exactly at a maturity of one year. Therefore, it is possible that our treatment sample contains commitments that were longer than 364 days and so do not receive a zero-risk weight. While this would bias against finding an effect from Basel I, we repeat our analysis on an alternative set of commitments that have maturities at origination strictly lower than one year.

One way to accomplish this objective is to exclude commitments with a reported maturity of 12 months. Effectively, we restrict the treatment sample to facilities that have a maturity less than or equal to 11 months. This guarantees that the treatment facilities benefited from the favorable capital regulation. The downside of this approach is that we exclude many facilities that benefited from the Basel I treatment. The results of this test are reported in panel B of Table 6. In this test, we use the shorter, balanced window around Basel I and the control group of commitments with 2- through 3-year maturities. As we can see from the negative sign and statistical significance of $BASEL1 \times ST$, we continue to

¹² DealScan has a variable with information on the maturity of the facility (which reports months to maturity) and another one with information on the type of the facility (which indicates whether it is a term loan, a credit line, a 364-day facility, and so forth). While nearly all facilities classified as 364-day have 12 months to maturity, many facilities have a maturity less than 12 months and benefit from the Basel I special treatment but are not classified as 364-day facilities in DealScan. In other words, exclusively relying on information about the maturity or the type of the credit facility could introduce some noise.

Table 6
Basel I: Tighter sample period and alternative maturity categorizations

<i>A. Narrow sample window</i>			
Variables	(1)	(2)	(3)
BASEL I	0.62 (0.61)	1.72 (1.32)	2.25 (1.64)
ST	-1.30 (-0.83)	-1.61 (-1.27)	-0.22 (-0.15)
BASEL I × ST	-4.80*** (-2.76)	-4.71*** (-3.04)	-6.16*** (-3.65)
Bank fixed effects	No	No	Yes
Observations	2,893	1,739	1,709
R-squared	.298	.383	.463
<i>B. Narrow window & ST defined as ≤ 11 months</i>			
Variables	(1)	(2)	(3)
BASEL I	0.51 (0.45)	1.69 (1.25)	1.85 (1.28)
ST	0.81 (0.46)	1.07 (0.53)	2.47 (1.02)
BASEL I × ST	-5.87** (-2.34)	-6.99** (-2.48)	-8.51*** (-2.90)
Bank fixed effects	No	No	Yes
Observations	2,401	1,434	1,402
R-squared	.211	.292	.380
<i>C. Narrow window & ST defined as ≤ 1 year pre-Basel I & 364-day post</i>			
Variables	(1)	(2)	(3)
BASEL I	0.32 (0.33)	1.45 (1.23)	1.78 (1.28)
364FAC	-1.27 (-0.81)	-1.61 (-1.26)	-0.15 (-0.10)
BASEL I × 364FAC	-6.88*** (-3.68)	-6.71*** (-3.95)	-8.34*** (-4.33)
Bank fixed effects	No	No	Yes
Observations	2,738	1,652	1,624
R-squared	.306	.396	.476

The dependent variable in each model is the annual undrawn fee on the credit line in basis points. Models include the same controls as the corresponding models in Table 3 including industry indicator variables. Panel A is restricted to credit lines with maturities up to 3 years and a sample period ranging from 3 years before and after Basel I (1990-1995). Panel B reports the results when short-term is defined as maturities up to 11 months. Panel C reports the results when short-term includes maturities up to (and including) 1 year before Basel I and 364-day facilities after Basel I. See the [Internet Appendix](#) for the definition of all the variables. Standard errors are clustered in two dimensions: borrower and time (year-quarter). * $p < .1$; ** $p < .05$; *** $p < .01$.

find that facilities with maturities strictly below 1 year experienced a reduction in undrawn fees relative to facilities with maturities up to 3 years.

Another way to account for this maturity categorization issue is to focus on facilities strictly designated as 364-day facilities by DealScan. This also poses a challenge because there were very few of these prior to Basel I. Indeed, there was no reason to adopt this specific maturity length or designation until

the Accord differentiated them. For this reason, we compare commitments with maturities up to and including one year issued prior to Basel I with specifically designated 364-day facilities taken out by borrowers afterward. The results of this test are reported in panel C of Table 6. Again, we use the shorter, balanced window around Basel I and a control group with maturities up to 3 years. Looking at the bottom panel of Table 6, we see that restricting our post-Basel I sample of short-term commitments to 364-day facilities does not affect our findings: we continue to observe that the relevant coefficient, $BASEL1 \times 364FAC$, is negative and statistically significant in each of our models.

In both cases, the more tailored treatment sample increases the magnitude of the interaction coefficient. This result is consistent with downward bias that would result from including untreated facilities in the short-term category. Compared to Table 4, panel B, which found a relative decline in undrawn fees of 4.7 to 6.2 bps, the restricted treatment samples find an impact of 5.9 to 8.5 bps, roughly 25% higher.

In Internet Appendix Table IA1, we repeat the robustness tests contained in Table 6 but allow the controls to vary in the pre- and post period (as in Table 4, panel B). These variations do not have a material impact on our conclusions thus far, as the key coefficient retains similar magnitude and statistical significance relative to earlier findings.

5.3 Placebo tests

The results presented above demonstrate the robustness of the relative decline in short-term commitment pricing under Basel I. However, one may wonder whether that reduction was driven by the discontinuity introduced by the Basel I Accord. In this section, we report the results of two placebo tests we design to confirm that our finding on undrawn fees is indeed driven by Basel I.

The first placebo test aims at addressing concerns that our results are driven by a general decline in the cost of short-term borrowing relative to long-term borrowing. To that end, we compare undrawn fees on commitments with maturities between 1 and 2 years with fees on commitments with maturities between 3 and 4 years. If changing capital regulations are the source of our results, we should not find a similar effect in this test because this sample of commitments received the same treatment under Basel I. If, on the other hand, our result is driven by a generalized decline in the relative cost of short-term commitment fees, then we should find some evidence of this decline among commitments with maturities between 1 and 2 years relative to longer maturity facilities.

Panel A of Table 7 reports the results of this investigation. As in previous robustness tests, we consider our narrow sample around Basel I. In this case, however, we include in the control group commitments with maturities between 3 and 4 years. In contrast to previous results, we find *no* evidence of a decline in the relative cost of 2-year maturity commitments. The coefficient of interest,

Table 7
Basel I: Placebo tests

<i>A. Undrawn fees on credit lines</i>			
Variables	(1)	(2)	(3)
BASEL1	−0.81 (−0.59)	1.03 (0.83)	0.26 (0.18)
ST2y	0.81 (0.50)	1.19 (0.85)	0.54 (0.47)
BASEL1 × ST2y	−0.01 (−0.00)	0.01 (0.01)	2.08 (1.06)
Bank fixed effects	No	No	Yes
Observations	2,640	1,568	1,418
R-squared	.202	.286	.387
<i>B. All-in-drawn spreads on credit lines</i>			
Variables	(1)	(2)	(3)
BASEL1	−7.72 (−0.88)	−4.67 (−0.56)	−7.77 (−0.97)
ST	10.22 (0.41)	29.50 (1.08)	17.36 (0.81)
BASEL1 × ST	−10.89 (−0.80)	−13.45 (−1.19)	−8.05 (−0.91)
Bank fixed effects	No	No	Yes
Observations	3,838	2,156	2,110
R-squared	.503	.613	.647
<i>C. All-in-drawn spreads on term loans</i>			
Variables	(1)	(2)	(3)
BASEL1	−36.35* (−1.79)	−54.86** (−2.00)	−65.86* (−1.96)
STtl	−7.06 (−0.54)	−20.45 (−1.12)	−3.77 (−0.15)
BASEL1 × STtl	12.49 (0.53)	14.53 (0.49)	7.88 (0.21)
Bank fixed effects	No	No	Yes
Observations	478	240	214
R-squared	.421	.566	.740

All dependent variables are in basis points. Each model is estimated on the sample of loans taken out 3 years before Basel I (1990-1992) and 3 years after (1993-1995). Panel A reports the results on undrawn fees for the sample of credit lines with maturities above 1 year and up to (and including) 4 years. *ST2y* indicates lines with a maturity equal to or less than 2 years. Panel B reports the results for all-in-drawn credit spreads over Libor for the sample of credit lines with maturities up to (and including) 3 years. Panel C reports the results for all-in-drawn credit spreads over Libor for the sample of term loans with maturities up to (and including) 3 years. Models include the same controls as the corresponding models in Table 3 including industry indicator variables. In addition, panels B and C use include the term premium (*TREASURY SLOPE*). Market controls (*BBBSPREAD* and *TREASURY SLOPE*) are allowed to vary between short- and long-term loans when we investigate all-in-drawn spreads in the last two panels. See the [Internet Appendix](#) for the definition of all the variables. Standard errors are clustered in two dimensions: borrower and time (year-quarter). * $p < .1$; ** $p < .05$; *** $p < .01$.

BASEL1 × *ST2y*, is not statistically significant in any of our models. In some of the models, this interaction term is even positive, although not significant. The lack of a decline suggests that there is not a general pattern in the pricing of commitments by maturity that would explain our primary finding.

The second placebo test builds on the all-in-drawn spreads of credit lines and term loans. Thus far, we have focused on the undrawn pricing component of credit lines because, as we noted above, we would expect Basel I to affect only undrawn fees and not credit spreads. Once drawn, credit lines receive the same treatment under Basel I regardless of their maturity. Recall that borrowers only pay the credit spread on the drawn portion of the facility.

To investigate that assertion, we estimate our pricing model, Equation 5, using credit lines' all-in-drawn spreads rather than their undrawn fees on the left-hand side. We use the all-in-drawn spreads, which includes fees in addition to the credit spread, because DealScan has comprehensive information on these spreads. Because the all-in-drawn spread contains some term risk, it is possible that banks consider the term premium in addition to the BBB spread when they consider the credit spread on these loans. For this reason we augment our market controls (M_t) to include the difference between the 10-year and 3-month treasuries at the time of the loan origination. Additionally, given that market conditions may affect the credit risk differently depending on the maturity of the loan, we allow our two market controls to vary between short- and long-term loans.

Panel B of Table 7 reports the results of this exercise. All-in-drawn spreads on short-term credit lines decline after Basel I by 8 to 13 bps relative to long-term commitments, depending on the model we consider, but this reduction is not statistically significant in any of our models. We find similar results when we consider refinements to isolate treated facilities similar to those we carried out to investigate undrawn fees.

In line with our assertion that Basel I should not affect the relative credit spreads, we also do not find that Basel I affected the relative all-in-drawn spreads of short term loans. We repeated our analysis of all-in-drawn spreads on term loans. In this case, we compare the change in cost of term loans with maturities up to 1 year relative to the cost of term loans with maturities between 1 and 3 years around Basel I. For term loans, we must rely on the all-in-drawn credit spread because borrowers do not pay an undrawn fee. Further, to reduce concerns with effects of joint pricing we exclude term loans in deals with credit lines affected by Basel I (we get similar results if we do not impose this condition). The results of this exercise, reported in panel C of Table 7, show that the interaction coefficient, $BASEL1 \times STtl$, is positive, but never statistically significant.

In sum, while we find strong evidence of a decline in the relative undrawn fees of commitments with maturities up to one year in the period immediately after Basel I, we do not find a similar effect on the relative credit spreads of either credit lines or term loans around that same period of time. This suggests that our evidence on undrawn fees of short-term credit lines is unlikely driven by a generalized decline in the relative cost of short-term funding and is instead the result of Basel I, which granted a special treatment to undrawn short-term credit lines.

5.4 Accounting for unobservables

Throughout, we have presented results from a model estimated with bank fixed effects (column 3). In the presence of bank fixed effects, identification comes from within bank comparisons between the pre- and post-Basel I periods. While we account for bank-specific controls, such as size, capital, and profitability, there may be unobserved changes in banks' pricing policies over time. One way to alleviate this concern is to use bank-year fixed effects, which allow pricing to vary for each bank by year.

Similarly, we have included borrower-specific controls; however, persistent, unobservable differences across borrowers could result in the pool of borrowers changing over time. Again, we can alleviate this concern by adding borrower fixed effects. This will focus on the smaller set of firms that are active borrowers in the loan market before and after Basel I. We also consider a specification that is even more restrictive: bank-borrower fixed effects. In this case the identification will derive from borrowers that take out credit lines before and after Basel I from the *same* bank.

The results of these tests are reported in [Internet Appendix Table IA2](#). In addition to the variation in fixed effects, we also consider the alternative categorizations of short-term lines. Model 1 reports the base categorization, which compares commitments with maturities up to (and including) 1 year with commitments with maturities between 1 and 3 years. Model 2 refines the previous analysis by leaving out from the target sample commitments with exactly 1-year maturity. Model 3 repeats the analysis we did before using 364-day facilities.

As one would expect, the more restrictive specifications yield smaller samples and weaker results from a statistical point of view. Nonetheless, both variables $BASEL1 \times ST$ and $BASEL1 \times 364FAC$ continue to be negative and of similar magnitude. The range of decline in short-term pricing is 5 to 10 bps. Further, these interaction terms continue to be statistically significant with the only exception (Model 2 in the bottom panel) possessing the smallest sample and just missing 10% statistical significance. Altogether, these results confirm that borrowers taking out short-term credit lines after Basel I benefited from a decline (relative to borrowers taking out longer-term credit lines) of about 6 bps on the undrawn fees on their credit lines when compared to that they were charged prior to Basel I.

5.5 Additional robustness tests

We conduct several additional robustness tests to our Basel I findings that are available upon request but are suppressed here for brevity. Thus far we have not continuously controlled for the maturity of the credit line, but rather have relied on a simple dummy variable to indicate short-term lines versus lines greater than one year. We reestimate our models including the log of the maturity of the credit line as a control variable and we find no meaningful impact on our findings.

We also consider variations of the placebo tests reported in Table 7 where we interact all of our controls with the *BASEL1* dummy variable and where we include the alternative fixed effects used in Section 5.4. These variations do not affect our conclusions. Lastly, throughout we assume the first year of the Basel I Accord is 1993, the year the Accord was fully phased-in for the United States. However, since U.S. banks were required to apply Basel I on a transitional basis starting in 1991, we have also done our tests using 1991 and 1992 as the first treatment year of the Accord. Additionally, we investigate what happens when we exclude credit lines originated close to the Basel I transition (1992 and 1993); thus, we compare loan pricing in 1990–1991 with loan pricing in 1994–1995. While these tests change some of our results, they do not change our key finding that commitments with maturities up to one year became relatively less costly following Basel I.

5.6 A test based on Basel II

We conduct a final test based on the changes implemented by the Basel II Accord. As we noted in Section 1, Basel II sought to restrict the unique treatment of short-term commitments granted by the Basel I. Basel II not only imposes additional conditions for credit lines to qualify for a reduced risk weight but also varied the resultant impact on regulatory capital based on whether the lending bank used the standardized approach or the advanced approach to determine capital requirements. As a consequence, fewer credit lines qualify for reduced capital holdings, significantly reducing the incentive to issue short-term lines rather than long-term lines. Further, those short-term lines that are issued could be less capital advantaged than the lines issued under Basel I, which would increase the costs to banks and, by extension, undrawn fees. While Basel II created additional obstacles for banks seeking to issue short-term credit lines that would qualify for a risk weight of zero, the heterogeneous implementation reduces our ability to draw strong inferences from Basel II.

A reason is that the implementation date of Basel II was subject to significant uncertainty, especially for U.S. banks. The Basel II Accord was finalized in June 2004, and U.S. federal agencies issued an advance notice of proposed rulemaking in August 2003, signaling their intent at the time to adopt Basel II. So, for the purpose of our analysis we assume 2004 implementation. However, the United States did not approve its implementation until November 2007.

As we see from Figure 1, the origination volume of 364-day facilities begins to decline in 2003, a decline that continues through 2006. The reversal effectively returns the share of short-term credit lines to pre-Basel I levels. This is true in both the level and share of originations. Looking at Figure 4, we also observe a change in relative pricing of undrawn fees. While undrawn fees of short-term commitments relative to long-term commitments decline dramatically after Basel I, we see the relative difference between the two shrink after Basel II, albeit by a smaller magnitude. Also, the post-Basel II decline

**Figure 4****Undrawn fees around Basel I and Basel II**

This figure plots the average annual undrawn fees on short- versus long-term credit lines around Basel I and Basel II. Fees are scaled to 100 in 1992 and 2004, respectively.

is associated with a decline in long-term credit line pricing rather than a clear increase in short-term credit line pricing. Hence the initial results are consistent with expectations: short-term issuance decreased, but the changes in pricing do not reverse from Basel I.

Despite the aforementioned caveats, we investigate the impact of Basel II on undrawn fees using our DiD approach (Equation 5). The results of this investigation are reported in Table 8. In the interest of space, we suppress all of the controls that are not critical to understanding the impact of Basel II on the relative pricing of credit lines with different maturities. In Section 4, we found that there was a decline in undrawn fees of commitments up to one year relative to those of longer term commitments after the passage of Basel I. Panel A of Table 8, which estimates a similar specification as Table 3, shows the reverse in response to Basel II; $BASEL2 \times ST$ is positive and statistically significant in each model. When we consider other specifications, such as those in Table 4, that tighten the sample (panel B) and allow for time-varying controls (panel C), the impact of Basel II on relative pricing of short-term lines is reduced and often not statistically significant. For example, the results in panel C with time-varying control coefficients show a relative price increase of only 2-3 bps and only marginally statistically significant in two of the three models we consider.

Following the approach we adopted to investigate unobserved bank and borrower factors around Basel I, we reestimate our pricing model with (a) bank-year, (b) bank and borrower, and (c) bank-borrower fixed effects. The results of

Table 8
Undrawn fees on credit lines around Basel II

<i>A. Baseline results</i>			
Variables	(1)	(2)	(3)
BASEL2	−6.35*** (−7.89)	−5.30*** (−6.76)	−5.21*** (−5.46)
ST	−5.77*** (−12.53)	−5.51*** (−9.09)	−5.54*** (−9.15)
BASEL2×ST	4.63*** (3.74)	4.12*** (5.24)	4.10*** (5.43)
Bank fixed effects	No	No	Yes
Observations	15,481	8,592	8,559
R-squared	.483	.595	.608
<i>B. Tighter sample (maturity ≤ 3 years)</i>			
Variables	(1)	(2)	(3)
BASEL2	−4.20*** (−5.55)	−4.82*** (−4.08)	−6.09*** (−4.17)
ST	−5.90*** (−11.01)	−6.01*** (−7.03)	−6.03*** (−7.00)
BASEL2×ST	1.92 (1.43)	3.00*** (3.23)	2.75** (2.62)
Bank fixed effects	No	No	Yes
Observations	7,239	4,099	4,064
R-squared	.438	.578	.595
<i>C. Tighter sample & time-varying coefficients</i>			
Variables	(1)	(2)	(3)
ST	−5.84*** (−9.38)	−5.94*** (−7.83)	−5.96*** (−7.72)
BASEL2×ST	3.12 (1.55)	2.81* (1.95)	2.68* (1.74)
Bank fixed effects	No	No	Yes
Observations	7,239	4,099	4,099
R-squared	.443	.586	.607

The dependent variable in each model is the annual undrawn fee on the credit line in basis points. Models include the same controls as the corresponding models in Table 3 including industry indicator variables. Panel A reports results based on the full sample of maturities ranging from 2000–2007 similar to Table 3. Panel B reports the results when we restrict the sample to credit lines with maturities up to 3 years similar to Table 4, panel A. Panel C reports results using the tighter control sample (maturity ≤ 3 years) and allowing for coefficients on controls to vary between the pre- and post-Basel II periods similar to Table 4, panel B. Because of lack of interpretability, we suppress the Basel I indicator coefficient in this panel. See the Internet Appendix for the definition of all the variables. Standard errors are clustered in two dimensions: borrower and time (year-quarter). * $p < .1$; ** $p < .05$; *** $p < .01$.

these tests are reported in the Internet Appendix IA3. When we add borrower fixed effects (panels B and C), the interaction terms that estimate the impact of Basel II on short-term credit line pricing lose their statistical significance and their magnitudes shrink. Along with Table 8, this suggests that our findings on the relative rise of undrawn fees on short-term commitments after Basel II are less robust than the Basel I results and may be driven by a change in the pool of borrowers rather than an increase in the fees on short-term commitments charged by banks. This finding also fits with our expectation that Basel II raised

the barrier for a credit to obtain a preferential risk weight, while retaining the possibility to receive a zero risk weight.

In sum, our robustness tests support our base findings that Basel I resulted in a statistically significant decline in undrawn fees for short-term lines, consistent with the idea that regulatory capital is costly for banks. We will investigate the economic significance of our estimates in the next section.

6. Economic Significance

The results thus far describe the statistical impact of the Basel I Accord. We have not yet interpreted the economic significance of the estimated magnitudes. In this section, we examine several dimensions by which the bank response to capital regulation affected the market for credit. First, we examine the impact of Basel I on borrowers' funding costs. Next, we examine whether the maturity composition of credit lines in the economy changed following Basel I. Finally, we use our loan pricing findings to infer the value of regulatory capital relief to banks.

6.1 Cost of corporate borrowing

One way to ascertain the economic significance of the pricing effects is to examine the benefit to the typical borrower. Our analysis finds that the decline in undrawn fees generally ranges from 5 to as high as 10 bps or roughly 10 to 25% of undrawn fees in the pre-Basel I period. Given the average size of a 364-day facility is \$600 million, the average undrawn facility will save a borrower approximately \$300 thousand to \$600 thousand per annum. However, the average facility is roughly one-third drawn reducing borrowers' savings to \$200 thousand to \$400 thousand per year.

6.2 Maturity of credit lines in the economy

A second measure of economic significance is whether the price changes induced a shift in the origination of short-term commitments. In other words, was the decline in the cost of short-term commitments induced by Basel I large enough to increase their relative importance in the total amount of credit lines banks grant corporations? Figure 1 displays a shift toward short-term commitments in both levels and market share and Table 9 reports the results of a more careful investigation.

We investigate issuance activity by estimating a DiD model to explain the share of issuance activity attributable to short-term commitments. Observations are at the bank-year-quarter level. Model 1 estimates whether the relative amount of credit lines with maturities up to one year increased in the years after Basel I. Models 2 through 4 investigate whether this effect is more pronounced among banks with low capital (those with a capital-to-asset ratio in the first quartile of the sample distribution). The sample period in Models 1 and 2 is the original timeframe around Basel I (1987 through 2003), whereas Model 3

Table 9
Basel I: Change in the share of short-term credit lines

Variables	(1)	(2)	(3)	(4)
BASEL1	0.07*** (2.77)	0.04 (1.22)	-0.08** (-2.00)	-0.09** (-2.51)
BASEL1 × LOWCAP		0.18*** (2.96)	0.18** (1.99)	0.17** (2.04)
LOWCAP		0.14 (0.76)	0.02 (0.07)	0.00 (0.01)
Constant	0.24*** (3.59)	0.17** (2.07)	0.47*** (3.06)	0.40** (2.34)
Bank controls	Yes	Yes	Yes	Yes
Observations	5,163	5,163	1,197	1,197
R-squared	.022	.037	.038	.064

The dependent variable is the ratio of the amount of credit lines originated with maturities up to one year and the total amount of credit lines the bank granted over the quarter. Models 1 and 2 are estimated on a sample of credit lines taken out between 1987 and 2003. Models 3 and 4 are estimated on a sample of credit lines taken out between 1990 and 1995. Model 3 considers only credit lines with maturities up to 3 years. Model 4 repeats Model 3 but with credit lines weighted by their maturity to account for the less frequent need to refinance. *LOWCAP* is a dummy variable for banks with an equity to assets ratio below the first quartile of the sample ratio (6.1%). All models include bank controls used in Table 3. Models 2 through 4 also include bank controls interacted with *LOWCAP*. See the [Internet Appendix](#) for the definition of all the variables. Standard errors are clustered in two dimensions: borrower and time (year-quarter). * $p < .1$; ** $p < .05$; *** $p < .01$.

restricts the analysis to credit lines taken out in the tighter window ranging from 1990 to 1995. Model 3 also considers an alternative ratio of short-term credit lines over the total amount of credit lines with maturities up to 3 years. Finally, in Model 4 we repeat Model 3 but we weight credit lines by their maturity to account for the less frequent need to refinance.

Table 9 shows two important results. First, the relative importance of short-term commitments increases in the period after the Basel I Accord (Model 1). Second, this increase is more prevalent among banks with low capital: *BASEL1* × *LOWCAP* is positive and significant in all of our models. These results are in line with the findings we unveiled in Table 6 showing that the decline in the price of short-term commitments was greater for low-capital banks. The issuance results demonstrate that the favorable treatment Basel I gave short-term commitments was sufficiently important to shift the maturity composition of corporate credit lines.

This shift towards short-term credit lines has several implications. For banks, it makes it easier for them to monitor borrowers and to manage the liquidity risk posed by credit lines. However, it exposes borrowers to additional refinancing risk as well as additional repricing risk. In fact, short maturity lines are significantly less likely to be a source of liquidity in distress ([Chodorow-Reich et al. 2022](#)). It is difficult to estimate the increase in refinancing risk that borrowers will experience when they shorten the maturity of their credit lines. However, the sensitivity of all-in-drawn spreads on one-year credit lines to the triple-B spread in the bond market is a good estimate of the additional repricing risk borrowers incur when they choose short term credit lines. Using a model similar to our Model 1 but restricting to credit lines

with maturities up to one year and using firm fixed effects, we find that a one-standard deviation increase in the triple-B spread in the bond market leads to an increase of about 14 bps in the all-in-drawn spread of triple B rated credit lines, the equivalent of a 20% increase when computed at the mean spread for these credit lines. This arguably exposes the firm to a meaningful increase in the cost of funding.

There are also potential macroeconomic risks to financial stability as market participants “optimize” their behavior to the presence of capital regulation. At the borrower level, short-maturity facilities become the predominant source of liquidity insurance, potentially undermining the efficacy of committed credit lines to weather prolonged disruptions. At the bank level, low-capital banks seem to capture the short-term facilities by underpricing their peers. Given that this behavior did not appear prior to Basel I, it suggests banks are willing to charge less for these facilities than might be justified solely by business conditions. In other words, low-capital banks’ aggressive response suggests the more fragile segment of the banking market takes on additional risk without commensurate returns. By extension, borrowers who come to rely on access to liquidity from banks may be at risk: both because of the capital that banks are able to forego and because of the concentration among already relatively less healthy issuers.

6.3 Cost of bank regulatory capital

Finally, our findings provide us with an opportunity to estimate how much banks are willing to pay to lower capital requirements and through this infer the value of regulatory capital to the banks, a metric that is intensely debated by the banking industry and policy makers. To do so, we use the framework outlined in Section 2.1.

Through the lens of the model, the difference-in-differences estimates can be used to infer the value of excess regulatory capital providing the bank is able to achieve an interior solution to its maximization problem. Recall, Equation 4 assumes banks maximize revenue taking prices as given; in other words, banks take demand for revolvers as given (for instance, the market for revolvers is competitive). Our base framework also simplifies the idea of regulatory constraints to a single definition of capital, rather than the various regulatory constraints banks can face. We consider the implications of relaxing each of these assumptions in the next section.

To use Equation 4 to infer the value placed on capital, we need to consider our estimated difference-in-differences estimates and the impact on required regulatory capital, $\bar{E}_{FL}$. Rearranging the equation and substituting for the marginal impact on required capital implies a post-Basel I equality condition,

$$\frac{(R_{FL} - R_{FS})_{Post} - (R_{FL} - R_{FS})_{Pre}}{\bar{E}_{FL}} = U'.$$

This ratio effectively summarizes the profits banks are willing to forego to avoid holding a marginal dollar of capital.¹³

The numerator is derived from our estimates on the relative change in cost of short-term versus long-term credit lines around Basel I. In the tighter sample period with time-varying controls these estimates were around 5 bps (Table 4, panel B). For low-capital banks in Table 5 panel B, the estimated change in pricing was as high as twice this number or 10 bps. Hence, the foregone annual profit for each dollar shifted to a short-term revolver is roughly 0.05 cents for the typical bank and 0.10 cents for a low-capital bank.

The denominator is calculated as the sensitivity of required capital to long-term credit facilities so we first must establish what type of capital we are considering. Basel I introduced two risk-weighted capital requirements: a tier 1 capital ratio and a total capital (tier 1 plus tier 2 capital) ratio.¹⁴ For this exercise we focus on required tier 1 capital ($\bar{E}$), as opposed to the total capital ratio, because tier 1 is the most stringent with respect to what qualifies as capital and is typically the most binding risk-weighted capital ratio. The minimum required tier 1 ratio was 4%. Under Basel I, an undrawn facility with a maturity shorter than one year received a risk weight of 0% while longer maturity facilities received a risk weight equal to the conversion factor of 50% times the risk weighting of a drawn commitment, 100%. It follows that the risk-weight of issuing a long-term facility is 50%, whereas the risk-weight for a short-term facility is 0%. The capital saved per dollar of risk weight reduction ($\bar{E}_{FL}$) is the product of 50% and the Tier 1 minimum of 4%. This implies that the shift of a \$1 undrawn commitment from a long-term revolver to a short-term facility reduces the required capital by \$0.02.¹⁵

In addition, we also need to consider that any change in profits will be subject to taxes. For this period we use the marginal corporate rate of 35%. Putting it all together we recover that the average bank was willing to give up 1.63 cents annually for every dollar of capital relief whereas low-capital banks were willing to forego twice that, or 3.25 cents annually per dollar of capital relief.

The analysis is admittedly back-of-the-envelope, but it reveals important cross-sectional differences in how banks value capital and highlights the critical assumptions necessary to infer the value of capital from a regulatory loophole. Given that banks' own estimates of regulatory capital costs are typically in double digits (e.g., 15%), our findings appear to be low, especially for the typical bank. But, the implied value of capital varies significantly in the cross-section, with low-capital banks willing to pay substantially more. In the context

¹³ This methodology is similar in spirit to Kisin and Manela (2016). See Anderson and Sallee (2011) for a use of this logic in the context of automobile regulation.

¹⁴ In addition, supervisors continued to consider the unweighted leverage ratio which is scaled by book assets. However, the leverage ratio did not change around Basel I and does not reflect off-balance-sheet commitments like revolvers.

¹⁵ $\bar{E} = RWA \times 4\%$, where $RWA = 50\% F^L + \omega L$, and ω is the risk-weight for other assets. It follows that the derivative of required capital w.r.t. long-term revolvers is $\bar{E}_{FL} = 50\% \times 4\% = 2\%$.

of other studies, our implied cost of capital is lower than recent estimates by [Benetton et al. \(2021\)](#) that were inferred during the financial crisis (10 to 16%), but greater than Kisin and Manela's (2016) estimates ($< 1\%$) derived from the years preceding the Global Financial Crisis. The estimates are roughly equivalent with the calculations in [Kashyap, Stein, and Hanson \(2010\)](#) that suggest a 25- to 45-bps increase in borrowing costs for a 10% change in capital requirements over a long-run time horizon. Our estimates also reflect a relatively healthy economic period, but the cross-sectional difference lend further support to the argument that the value of capital varies with the state of the bank.

With respect to critical assumptions, several are worth discussing. First, the difference-in-differences compares credit lines over time and relies on a linear estimation specification and numerous controls to isolate the differences in pricing arising from changes in capital rules. With respect to the model, our base scenario assumes a competitive market. In other words, banks are price takers. If competition is imperfect for credit lines, prices will be set higher for both long and short-term lines as banks internalize their impact on market prices. In such a scenario, the relative demand elasticities between short and long-term lines will affect the interpretation of our estimates. For example, if the demand elasticity for long-term lines is greater than or equal to short-term lines we are underestimating the value of capital as banks will mark up short-term lines more so than long-term lines in an effort to maximize revenue.¹⁶

Similar to the imperfect competition case, constraints on the quantity of originations will limit the response of prices. In the most likely scenario that the demand for short-term lines is limited, prices will not fall as far as they might otherwise in response to Basel I, again suggesting that we are recovering a lower bound on the willingness of banks to pay for capital relief.

Finally, if banks consider alternative capital constraints, this will also impact the estimates. A binding leverage constraint should not affect what we recover because it does not apply to either short-term or long-term undrawn lines. But, if banks target a specific level of a risk-weighted capital ratio rather than a capital buffer over a regulatory minimum or if banks target the total capital ratio rather than the tier 1 capital ratio, it would lower our estimates with respect to the value of capital suggesting we are recovering an upper bound.¹⁷ Nonetheless, we believe our base assumption that banks target a buffer to the tier 1 capital ratio is supported by the literature (e.g., [Berger et al. 2008](#)).

¹⁶ A more complicated case would be the presence of differential movements in pass-through across short- and long-term lines over time. We have no reason to suspect that pass-through is different across maturity or that this differential is changing around Basel I; however, if this assumption is incorrect it would confound our empirical strategy and make the interpretation ambiguous.

¹⁷ Rather than the marginal impact of equity being equal to the risk-weight times the regulatory tier 1 minimum of 4% it would be multiplied by the relevant ratio. For example, in the case of the total capital ratio that would be 8%.

Importantly, the cross-sectional insight from our results remains relevant in these various settings.

7. Final Remarks

In this paper, we exploit a discontinuity introduced by Basel I in the capital treatment of undrawn commitments with maturities up to one year. We find strong statistical evidence that bank pricing behavior reflects that regulatory capital is costly for banks. Undrawn fees of commitments with maturity less than one year decline relative to longer term commitments in the years immediately after the implementation of Basel I. Our findings are robust to a wide array of methods and samples. The results do not appear to be driven by other facets of the Basel I Accord or time-varying market conditions because our results are stronger among low-capital banks and we do not find similar evidence in placebo tests. Further, and as expected, we find a reversal in short-term originations following Basel II, which reduced the scope of the favorable treatment under Basel I.

We link the difference-in-differences estimates to the bank's problem to show that banks are willing to pay almost \$0.02 cents for a dollar saving in their capital buffer and that low-capital banks are willing to pay roughly twice that amount. While below what banks suggest is the cost of capital, the change in pricing was enough for lenders to induce a significant transformation in the composition of credit lines in the marketplace. During the Basel I period there was a large shift towards shorter-term credit lines that appears primarily explained by their regulatory treatment, hence it may be that some banks were willing to pay even more but that they had exhausted their opportunities to convert borrowers.

Finally, our paper has some important insights for the design of regulation. First, our evidence on banks' adjustments to credit lines' pricing confirms that discontinuous treatment of "similar" securities induces regulatory optimization. Second, our evidence on the rapid growth of 364-day facilities upon the introduction of Basel I and the equally rapid decline in these contracts after Basel II illustrates the sensitivity of the marketplace to regulatory changes. Finally, our paper shows a novel link between capital regulation and liquidity risk. By offering a differential treatment to commitments with different maturities that affect their relative cost, capital regulation can alter the maturity preferences of corporate borrowers and consequently the liquidity risk they pose to banks.

References

- Anderson, S. T., and J. M. Sallee. 2011. Using loopholes to reveal the marginal cost of regulation: The case of fuel-economy standards. *American Economic Review* 101:1375–409.
- Beatty, A., and A. Gron. 2001. Capital, Portfolio, and Growth: Bank Behavior Under Risk-Based Capital Guidelines. *Journal of Financial Services Research* 20:5–31.

- Behn, M., R. Haselmann, and P. Wachtel. 2016. Procyclical Capital Regulation and Lending. *Journal of Finance* 71:919–56.
- Benetton, M., P. Eckley, N. Garbarino, L. Kirwin, and G. Latsi. 2021. Capital requirements and mortgage pricing: Evidence from Basel II. *Journal of Financial Intermediation* 48.
- Berger, A. N., R. DeYoung, M. J. Flannery, D. Lee, and Ö. Öztekin. 2008. How do large banking organizations manage their capital ratios? *Journal of Financial Services Research* 34:123–49.
- Berger, A. N., and G. F. Udell. 1993. Did risk-based capital allocate bank credit and cause a ‘credit crunch’ in the U.S.? *Social Science Research Network*.
- Bernanke, B., and C. S. Lown. 1991. The credit crunch. *Brookings Papers on Economic Activity* 22:205–48.
- Bord, V. M., and J. Santos. 2014. Banks’ liquidity and the cost of liquidity to corporations. *Journal of Money, Credit and Banking* 46:13–45.
- Burhouse, S., J. Feid, G. French, and K. Ligon. 2003. Basel and the evolution of capital regulation: Moving forward, looking back. *An Update on Emerging Issues in Banking*.
- Chodorow-Reich, G., O. Darmouni, S. Luck, and M. Plosser. 2022. Bank liquidity provision across the firm size distribution. *Journal of Financial Economics* 144:908–32.
- Cornett, M. M., J. J. McNutt, P. E. Strahan, and H. Tehranian. 2011. Liquidity risk management and credit supply in the financial crisis. *Journal of Financial Economics* 101:297–312.
- Fama, E. 1998. Market efficiency, long-term returns, and behavioral finance. *Journal of Financial Economics* 49:283–306.
- Gropp, R., T. Mosk, S. Ongena, and C. Wix. 2019. Banks response to higher capital requirements: Evidence from a quasi-natural experiment. *Review of Financial Studies* 32:266–99.
- Hancock, D., and J. A. Wilcox. 1993. Has there been a “capital crunch” in banking? the effects on bank lending of real estate market conditions and bank capital shortfalls. *Journal of Housing Economics* 3:31–50.
- . 1998. The “credit crunch” and the availability of credit to small business. *Journal of Banking & Finance* 22:983–1014.
- Haubrich, J., and P. Wachtel. 1993. Capital requirements and shifts in commercial bank portfolios. *Economic Review* 29:2–15.
- Ivashina, V., and D. Scharfstein. 2010. Bank lending during the financial crisis of 2008. *Journal of Financial Economics* 97:319–38.
- Jiménez, G., S. Ongena, J.-L. Peydró, and J. Saurina. 2017. Macroprudential policy, countercyclical bank capital buffers and credit supply: evidence from the spanish dynamic provisioning experiments. *Journal of Political Economy* 125:2126–77.
- Kashyap, A. K., J. C. Stein, and S. G. Hanson. 2010. An analysis of the impact of ‘substantially heightened’ capital requirements on large financial institutions.
- Kisin, R., and A. Manela. 2016. The Shadow Cost of Bank Capital Requirements. *The Review of Financial Studies* 29:1780–820.
- Modigliani, F., and M. H. Miller. 1958. The cost of capital, corporation finance and the theory of investment. *The American Economic Review* 48:261–97.
- Peek, J., and E. Rosengren. 1995. Bank regulation and the credit crunch. *Journal of Banking & Finance* 19:679–92.
- Pennacchi, G. G., and J. A. Santos. 2021. Why do banks target roe? *Journal of Financial Stability* 54:S1572308921000152–.
- Santos, J. 2011. Bank corporate loan pricing following the subprime crisis. *Review of Financial Studies* 24:1916–43.

Santos, J. A. C. 2001. Bank capital regulation in contemporary banking theory: A review of the literature. *Financial Markets, Institutions & Instruments* 10:41–84.

Santos, J. A. C., and S. V. Viswanathan. 2020. Bank Syndicates and Liquidity Provision. Working Paper 27701, Federal Reserve Bank of New York.

Van den Heuvel, S. J. 2008. The welfare cost of bank capital requirements. *Journal of Monetary Economics* 55:298–320.